

DANDI:000409: IBL - Brain Wide Map

Scope of this export (computed by this export): only the 459 desc-processed_behavior+ecephys.nwb files were used (spike-sorted units, trials and behaviour, ~0.5 TB). The raw electrophysiology and video files (~49 TB) were not downloaded.

Source labels: **DANDI metadata** = the Dandiset's published metadata; **NWB file** = values or text stored in the NWB files; **Computed by this export** = derived by the export scripts in dandi_export / (definitions given where used).

1. The dataset as published

Everything in this section is quoted from the Dandiset's published metadata or, for file-level fields, from sub-SWC-066_ses-a4747ac8-6a75-444f-b99b-696fff0243fd_desc-processed_behavior+ecephys.nwb.

Field	Value (DANDI metadata)
Name	IBL - Brain Wide Map
Identifier / version	DANDI:000409 / 0.260309.1324
Citation	International Brain Laboratory; Benson, Brandon; Benson, Julius; Birman, Daniel; Bonacchi, Niccolò; Carandini, Matteo; Catarino, Joana; Chapuis, Gaelle; Dayan, Peter; DeWitt, Eric; Engel, Tatiana; Fabbri, Michele; Faulkner, Mayo; Fiete, Ila; Findling, Charles; Freitas-Silva, Laura; Gerçek, Berk; Harris, Kenneth; Hofer, Sonja; Hu, Fei; Hubert, Félix; Huntenburg, Julia; Khanal, Anup; Langdon, Christopher; Lau, Petrina; Meijer, Guido; Miska, Nathaniel; Noel, Jean-Paul; Nylund, Kai; Pan-Vazquez, Alejandro; Pouget, Alexandre; Rossant, Cyrille; Roth, Noam; Schaeffer, Rylan; Schartner, Michael; Shi, Yanliang; Socha, Karolina; Steinmetz, Nicholas; Svoboda, Karel; Urai, Anne; Wells, Miles; West, Steven; Whiteway, Mathew; Winter, Olivier; Witten, Ilana (2026) IBL - Brain Wide Map (Version 0.260309.1324) [Data set]. DANDI Archive. https://doi.org/10.48324/dandi.000409/0.260309.1324
DOI	10.48324/dandi.000409/0.260309.1324
Contributors (roles)	International Brain Laboratory (ContactPerson); Wellcome Trust (Funder); Simons Foundation (Funder); National Institutes of Health (Funder); Benson, Brandon; Benson, Julius; Birman, Daniel; Bonacchi, Niccolò; Carandini, Matteo; Catarino, Joana; Chapuis, Gaelle; Dayan, Peter; DeWitt, Eric; Engel, Tatiana; Fabbri, Michele; Faulkner, Mayo; Fiete, Ila; Findling, Charles; Freitas-Silva, Laura; Gerçek, Berk; Harris, Kenneth; Hofer, Sonja; Hu, Fei; Hubert, Félix; Huntenburg, Julia; Khanal, Anup; Langdon, Christopher; Lau, Petrina; Meijer, Guido; Miska, Nathaniel; Noel, Jean-Paul; Nylund, Kai; Pan-Vazquez, Alejandro; Pouget, Alexandre; Rossant, Cyrille; Roth, Noam; Schaeffer, Rylan; Schartner, Michael; Shi, Yanliang; Socha, Karolina; Steinmetz, Nicholas; Svoboda, Karel; Urai, Anne; Wells, Miles; West, Steven; Whiteway, Mathew; Winter, Olivier; Witten, Ilana
License	CC-BY-4.0
Keywords	International Brain Laboratory, multi-probe, Neuropixels, Allen Mouse Brain CCFv3, Allen Mouse Brain Atlas, head-fixed, decision-making, face-tracking
Related resources	A Brain-Wide Map of Neural Activity during Complex Behaviour https://doi.org/10.1101/2023.07.04.547681 (IsDescribedBy); Data release - Brainwide map - Q4 2022 https://figshare.com/articles/preprint/Data_release_-_Brainwide_map_-_Q4_2022/21400815 (IsDescribedBy); Behavior: Appendix 2: IBL protocol for mice training https://doi.org/10.1101/2020.01.17.909838 (IsDescribedBy); Conversion Software https://github.com/catalystneuro/IBL-to-nwb (IsSourceOf)
Files / size / subjects	2048 files, 49,700.60 GB, 139 subjects
Species	Mus musculus - House mouse
Approach	electrophysiological approach; behavioral approach
Measurement technique	surgical technique; multi electrode extracellular electrophysiology recording technique; analytical technique; behavioral technique; spike sorting technique
Variables measured	ElectricalSeries, ElectrodeGroup, SpatialSeries, Units, ProcessingModule
Date published	2026-03-09T13:24:10.741911+00:00

Description (DANDI metadata, verbatim)

The International Brain lab (IBL) aims to understand the neural basis of decision-making in the mouse by gathering a whole-brain activity map composed of electrophysiological recordings pooled from multiple laboratories. We have systematically recorded from nearly all major brain areas with Neuropixels probes, using a grid system for unbiased sampling and replicating each recording site in at least two laboratories. These data have been used to construct a brain-wide map of activity at single-spike cellular resolution during a decision-making task. In addition to the map, this data set contains other information gathered during the task: sensory stimuli presented to the mouse; mouse decisions and response times; and mouse pose information from video recordings and DeepLabCut analysis.

File-level fields (NWB file, verbatim)

Field	Value (NWB file)
institution	Sainsbury Wellcome Centre
lab	hoferlab
protocol	_iblrig_tasks_ephysChoiceWorld6.5.3
subject	subject_id: SWC_066; species: Mus musculus; sex: F
devices	Probe00: A PRB_1_4_0480_1_C probe. Additional metadata: {"model_name": "PRB_1_4_0480_1_C", "manufacturer": "imec", "port": "1", "slot": "3", "part_number": "PRB_1_4_0480_1_C"}
electrode_groups	Probe00: description = 'Electrode group for probe00', location = 'Unresolved'

2. Contents of the processed files

Quantity	Value	Source
Sessions / subjects / labs	459 / 139 / 12	NWB file
Sessions per lab (NWB lab field)	hausserlab (62), danlab (46), mainenlab (44), cortexlab (42), churchlandlab_ucla (41), angelakilab (41), zadorlab (37), wittenlab (37), churchlandlab (36), steinmetzlab (31), mrsicflogellab (25), hoferlab (17)	NWB file
Sex, by subject	M: 94, F: 45	NWB file
Age at session (session start – subject.date_of_birth)	median 182 days (range 92–461)	Computed by this export
Probes per session	1: 219 sessions, 2: 240 sessions	NWB file
Probe device description (example file)	Probe00: A PRB_1_4_0480_1_C probe. Additional metadata: {"model_name": "PRB_1_4_0480_1_C", "manufacturer": "imec", "port": "1", "slot": "3", "part_number": "PRB_1_4_0480_1_C"}	NWB file
Task protocol(s) (NWB protocol field)	_iblrig_tasks_ephysChoiceWorld; _iblrig_tasks_ephysChoiceWorld6.1.3; _iblrig_tasks_ephysChoiceWorld6.2.3; _iblrig_tasks_ephysChoiceWorld6.2.5; _iblrig_tasks_ephysChoiceWorld6.4.0; _iblrig_tasks_ephysChoiceWorld6.4.1; _iblrig_tasks_ephysChoiceWorld6.4.2; _iblrig_tasks_ephysChoiceWorld6.4.3; _iblrig_tasks_ephysChoiceWorld6.5.0; _iblrig_tasks_ephysChoiceWorld6.5.3; _iblrig_tasks_ephysChoiceWorld6.6.1; _iblrig_tasks_ephysChoiceWorld6.6.2; _iblrig_tasks_ephysChoiceWorld6.6.4	NWB file
Trials	296,090	NWB file
gabor_stimulus_contrast values (%)	0, 6.25, 12.5, 25, 100	NWB file
probability_left values	0.2, 0.5, 0.8	NWB file
mouse_wheel_choice values	clockwise (148,816), counter_clockwise (146,122), none (1,152)	NWB file
quiescence_period range	0.40–0.70 s (median 0.52 s)	NWB file
Units (ibl_quality_score = 1)	621,733 (75,708)	NWB file
ibl_quality_score counts	0.00: 112,878, 0.33: 255,535, 0.67: 177,612, 1.00: 75,708	NWB file
kilosort2_label counts	mua: 472,584, good: 149,149	NWB file
Distinct electrode location labels at units' max_electrode	579	NWB file
Total spikes (sum of spike_count)	21,140,973,555	NWB file

Definitions of ibl_quality_score, the trial columns and the electrode columns are quoted in section 8.

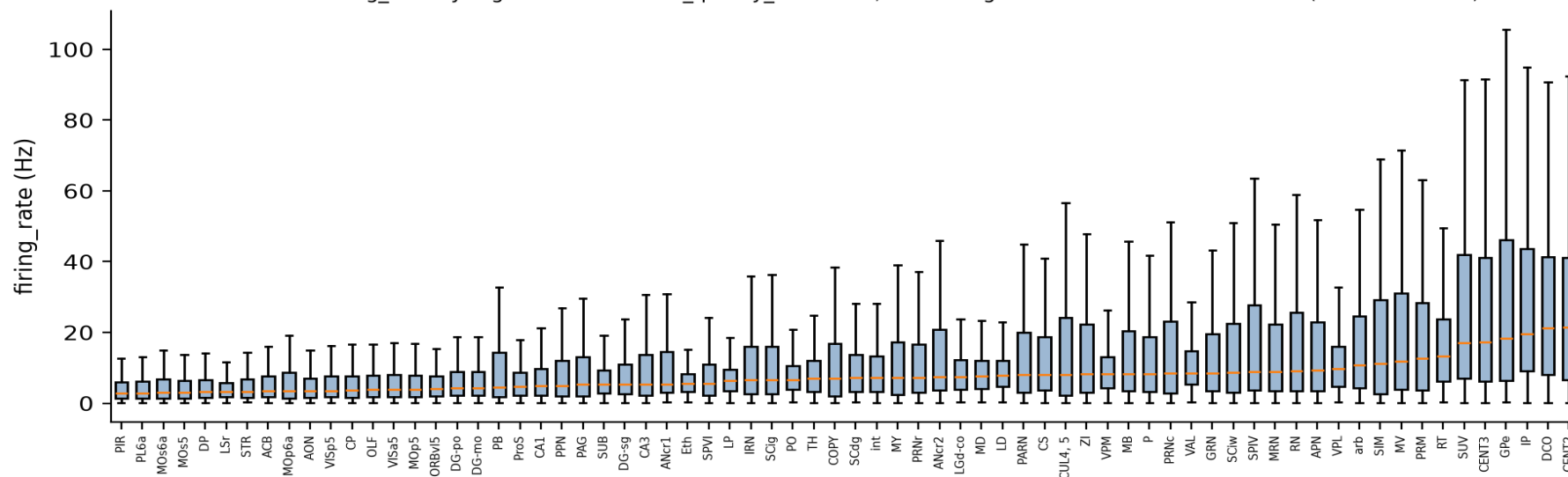
3. Export parameters

Parameter	Definition	Source
region_name	electrodes.location at the unit's max_electrode	NWB file
region	Allen acronym matched to region_name (punctuation-insensitive) using allen_structures.csv, a snapshot of the Allen Mouse Brain Atlas ontology	Computed by this export
good	ibl_quality_score = 1.0 (the NWB description: '1.0 = all three passed')	NWB file
firing_rate_hz	units.firing_rate as stored in the NWB file	NWB file
peak_to_trough_ms	units.peak_to_trough_duration_ms (values of –1 treated as missing)	NWB file

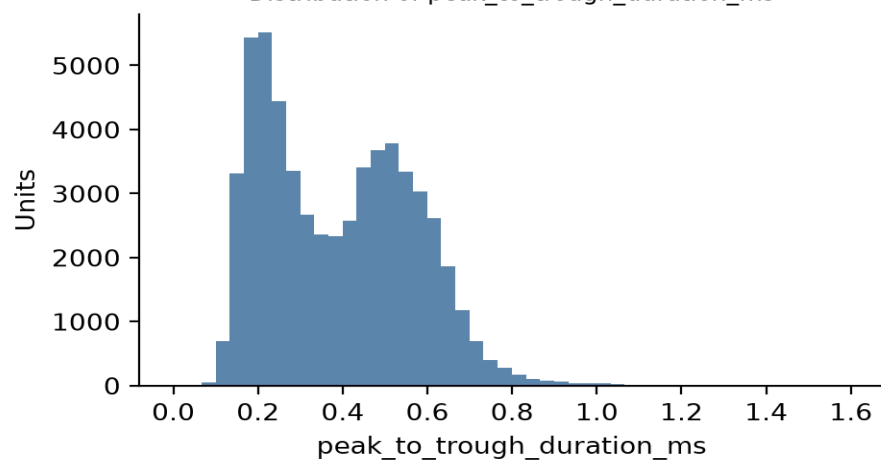
Parameter	Definition	Source
task_rate_hz	spikes between first trial start and last trial stop / that duration	Computed by this export
prestim_rate_hz, stim_rate_hz	mean over trials of count_prestim / 0.3 s and count_stim / 0.25 s	Computed by this export
count_prestim	spikes from -0.4 to -0.1 s relative to trials.gabor_stimulus_onset_time	Computed by this export
count_stim	spikes from +0 to +0.25 s relative to trials.gabor_stimulus_onset_time	Computed by this export
count_remove	spikes from -0.25 to +0 s relative to trials.wheel_movement_onset_time	Computed by this export
count_feedback	spikes from +0 to +0.5 s relative to trials.feedback_time	Computed by this export

4. Overview figures (Computed by this export)

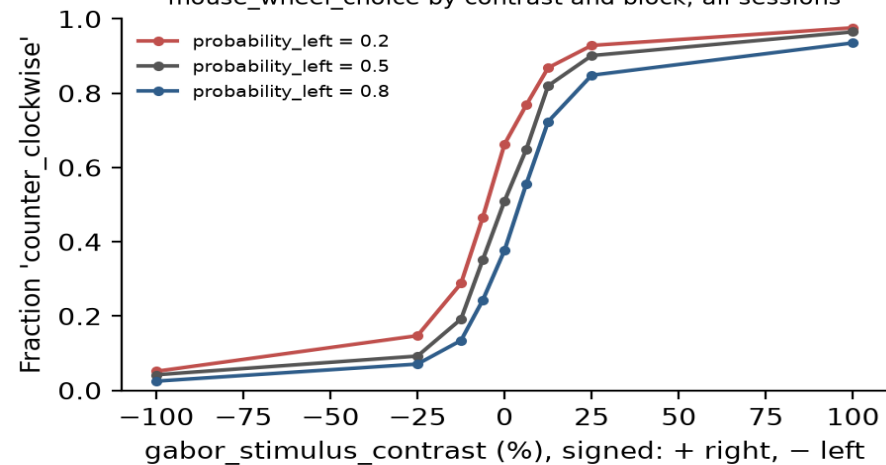
NWB units.firing_rate by region: units with ibl_quality_score = 1, the 70 regions with the most such units (outliers hidden)



Distribution of peak_to_trough_duration_ms



mouse_wheel_choice by contrast and block, all sessions



5. All regions (Computed by this export)

One row per region for units with ibl_quality_score = 1, excluding unassigned labels ('root', 'void', or names not found in the Allen ontology). Sorted by number of units. firing_rate is the NWB value; pre-stimulus and stimulus rates are computed by this export (section 3).

Acronym	Region (NWB location)	Units	Sessions	Mice	firing_rate mean (Hz)	SD (Hz)	Median (Hz)	Pre-stim mean (Hz)	Pre-stim median (Hz)	Stim mean (Hz)	peak_to_trough median (ms)
CP	Caudoputamen	2892	71	42	6.70	11.35	3.62	5.89	2.67	7.61	0.43
MRN	Midbrain reticular nucleus	2678	110	71	16.03	18.05	8.95	15.56	7.74	17.80	0.23
PO	Posterior complex of the thalamus	2329	50	44	8.30	7.37	6.65	7.81	5.72	9.30	0.47
LP	Lateral posterior nucleus of the thalamus	1812	72	62	7.35	6.09	6.30	6.12	4.65	7.34	0.47
CA1	Field CA1	1807	133	91	7.44	7.89	4.86	6.52	4.20	8.14	0.40
arb	arbor vitae	1684	71	40	18.82	22.82	10.76	18.25	9.62	19.71	0.20
APN	Anterior pretectal nucleus	1543	51	42	15.98	16.93	9.34	15.31	7.96	17.13	0.20
MB	Midbrain	1465	136	79	14.94	17.02	8.24	13.90	6.85	16.04	0.23
VPM	Ventral posteromedial nucleus of the thalamus	1310	37	32	10.06	8.79	8.22	9.25	7.15	11.68	0.40
CUL4, 5	Lobules IV-V	1207	32	26	19.85	29.34	8.11	19.12	6.50	20.79	0.27
LSr	Lateral septal nucleus rostral (rostroventral) part	1156	29	21	4.81	5.79	3.22	4.45	3.02	5.07	0.47
CA3	Field CA3	950	75	56	10.13	11.83	5.46	9.92	5.13	11.02	0.37
SUB	Subiculum	914	34	30	7.35	7.32	5.31	5.43	3.72	7.38	0.50
SIM	Simple lobule	893	21	16	25.13	34.88	11.31	24.74	8.46	26.33	0.23
PRNr	Pontine reticular nucleus	770	26	18	12.24	13.76	7.34	10.79	4.98	12.98	0.17
P	Pons	751	43	32	15.54	19.68	8.33	15.27	6.87	17.75	0.20
IRN	Intermediate reticular nucleus	744	31	21	12.34	16.22	6.61	11.94	4.72	15.30	0.17
MOp5	Primary motor area Layer 5	691	27	18	6.02	6.52	3.97	5.61	3.28	7.10	0.53
MV	Medial vestibular nucleus	672	22	18	22.74	27.20	11.85	22.72	9.38	22.35	0.20
SCig	Superior colliculus motor related intermediate gray layer	671	51	34	11.90	14.13	6.65	11.12	5.06	12.73	0.23
CENT3	Lobule III	663	27	22	29.81	33.55	17.38	28.81	14.66	29.97	0.20
MOs5	Secondary motor area layer 5	661	33	25	4.71	5.12	3.10	4.35	2.63	6.08	0.57
SCiw	Superior colliculus motor related intermediate white layer	655	46	31	16.20	19.09	8.67	14.81	7.03	16.84	0.20
PAG	Periaqueductal gray	630	37	25	10.38	14.04	5.25	9.44	3.94	11.33	0.27
IP	Interposed nucleus	619	28	21	30.94	31.58	19.54	31.82	19.51	30.03	0.17
PIR	Piriform area	614	29	21	4.96	6.09	2.79	4.86	2.65	4.81	0.50
CENT2	Lobule II	613	14	11	29.17	29.06	21.46	28.45	19.03	29.68	0.20
MD	Mediodorsal nucleus of thalamus	604	21	16	9.16	6.92	7.62	8.91	7.13	11.58	0.50
LD	Lateral dorsal nucleus of thalamus	581	22	19	9.79	9.00	7.87	8.06	5.99	10.64	0.47
GRN	Gigantocellular reticular nucleus	560	17	13	13.74	14.88	8.59	12.65	6.20	19.36	0.20
STR	Striatum	526	56	38	6.26	10.75	3.25	5.88	2.84	6.27	0.47
ANcr2	Crus 2	519	15	10	21.39	35.79	7.43	21.60	6.64	23.00	0.33
MOp6a	Primary motor area Layer 6a	499	25	17	6.25	7.60	3.46	4.90	2.50	7.52	0.53
VIa5	Anterior area layer 5	494	35	34	5.89	5.91	3.92	4.75	2.88	5.98	0.53
int	internal capsule	465	26	22	11.25	14.22	7.18	10.79	6.33	12.95	0.23
MOs6a	Secondary motor area layer 6a	463	31	26	5.22	7.44	3.04	4.42	2.07	6.60	0.53

Acronym	Region (NWB location)	Units	Sessions	Mice	firing_rate mean (Hz)	SD (Hz)	Median (Hz)	Pre-stim mean (Hz)	Pre-stim median (Hz)	Stim mean (Hz)	peak_to_tro ugh median (ms)
DG-sg	Dentate gyrus granule cell layer	442	92	75	7.99	7.88	5.35	7.56	4.58	8.92	0.37
VPL	Ventral posterolateral nucleus of the thalamus	436	23	22	11.60	9.21	9.84	9.93	7.53	13.90	0.40
RN	Red nucleus	429	22	20	17.50	19.25	9.02	16.84	8.93	19.09	0.20
AON	Anterior olfactory nucleus	428	15	13	5.08	4.99	3.51	4.40	2.55	4.89	0.53
PARN	Parvicellular reticular nucleus	423	21	17	15.16	18.10	8.04	14.49	6.00	15.50	0.17
ANcr1	Crus 1	421	10	8	13.85	22.60	5.46	13.49	4.88	15.41	0.30
RT	Reticular nucleus of the thalamus	421	25	19	16.78	13.69	13.39	14.33	9.87	20.53	0.23
OLF	Olfactory areas	412	36	24	6.55	8.00	3.82	5.99	3.34	6.07	0.47
MY	Medulla	397	53	34	12.74	15.81	7.33	12.66	4.81	13.36	0.17
CS	Superior central nucleus raphe	391	10	8	13.16	13.49	8.09	12.71	6.85	13.93	0.23
PRM	Paramedian lobule	382	15	9	25.23	34.89	12.62	24.12	8.78	28.15	0.27
ProS	Prosubiculum	378	22	22	6.71	6.89	4.71	5.88	4.16	6.82	0.50
VISp5	Primary visual area layer 5	374	35	27	5.35	5.18	3.51	4.47	2.79	5.47	0.50
LGd-co	Dorsal part of the lateral geniculate complex core	370	22	18	8.79	6.32	7.49	7.56	6.12	9.03	0.40
PPN	Pedunclopontine nucleus	369	19	15	9.54	11.97	4.92	9.57	4.02	10.68	0.23
Eth	Ethmoid nucleus of the thalamus	366	20	19	7.21	7.56	5.51	6.39	4.29	7.99	0.50
DG-po	Dentate gyrus polymorph layer	366	62	55	6.52	6.35	4.35	6.81	4.29	7.30	0.43
ACB	Nucleus accumbens	340	23	17	7.40	10.94	3.45	7.09	3.08	8.10	0.50
DG-mo	Dentate gyrus molecular layer	329	82	66	7.71	9.91	4.37	6.92	3.47	7.94	0.40
SPIV	Spinal vestibular nucleus	327	17	12	20.45	26.13	8.95	20.50	9.10	20.72	0.17
GPe	Globus pallidus external segment	322	13	9	31.01	31.83	18.28	31.19	17.91	32.58	0.23
SPVI	Spinal nucleus of the trigeminal interpolar part	321	15	10	8.70	9.74	5.60	8.55	4.33	10.47	0.17
DP	Dorsal peduncular area	318	14	11	5.04	6.01	3.18	4.41	2.85	4.85	0.53
DCO	Dorsal cochlear nucleus	312	13	12	26.64	23.03	21.32	26.24	19.30	26.60	0.23
ORBvl5	Orbital area ventrolateral part layer 5	310	5	5	6.12	8.05	4.18	5.19	3.00	6.10	0.57
TH	Thalamus	310	57	47	9.61	9.24	6.96	8.52	6.03	10.72	0.40
ZI	Zona incerta	296	40	30	14.73	16.42	8.19	13.50	8.26	16.66	0.23
VAL	Ventral anterior-lateral complex of the thalamus	295	10	10	12.58	12.41	8.52	11.15	6.88	15.68	0.43
SCdg	Superior colliculus motor related deep gray layer	285	34	25	11.50	12.89	7.16	11.09	5.93	13.16	0.20
COPY	Copula pyramidis	284	13	9	15.72	24.57	7.00	15.16	5.03	16.31	0.27
PRNc	Pontine reticular nucleus caudal part	280	11	9	15.91	18.46	8.49	15.45	6.26	19.05	0.20
PB	Parabrachial nucleus	278	25	21	13.85	22.32	4.54	13.45	3.72	14.02	0.22
SUV	Superior vestibular nucleus	277	13	12	28.97	30.24	17.12	28.33	14.26	27.07	0.17
PL6a	Prelimbic area layer 6a	273	13	12	4.79	6.13	2.89	3.93	2.10	6.18	0.57
TTd	Taenia tecta dorsal part	270	12	11	5.83	7.45	3.43	5.07	3.43	5.38	0.53
ORBvl6a	Orbital area ventrolateral part layer 6a	259	6	6	4.57	6.32	2.40	3.62	1.61	4.34	0.60
SI	Substantia innominata	256	27	20	14.66	15.20	8.79	14.65	8.66	14.56	0.33
ENTl5	Entorhinal area lateral part layer 5	255	14	11	4.11	5.39	2.21	3.83	1.95	4.24	0.57
ViSa6a	Anterior area layer 6a	255	33	32	5.37	5.76	3.68	3.92	1.92	6.12	0.47
UVU	Uvula (IX)	245	8	5	18.67	29.77	7.75	18.32	4.89	18.09	0.23
ICe	Inferior colliculus external nucleus	237	17	16	18.91	25.17	11.82	17.88	8.91	20.25	0.20

Acronym	Region (NWB location)	Units	Sessions	Mice	firing_rate mean (Hz)	SD (Hz)	Median (Hz)	Pre-stim mean (Hz)	Pre-stim median (Hz)	Stim mean (Hz)	peak_to_tro ugh median (ms)
VISp6a	Primary visual area layer 6a	235	38	28	6.45	7.67	3.77	5.33	2.73	6.90	0.43
CB	Cerebellum	235	34	26	20.45	22.96	10.58	19.52	10.47	21.03	0.20
MOp2/3	Primary motor area Layer 2/3	234	20	15	5.20	5.66	3.06	4.56	2.48	5.50	0.57
POST	Postsubiculum	233	18	17	10.03	12.51	5.84	7.54	3.14	9.81	0.30
ENTm3	Entorhinal area medial part dorsal zone layer 3	230	8	8	5.64	5.80	4.15	4.88	3.70	5.62	0.57
PoT	Posterior triangular thalamic nucleus	228	19	19	9.19	9.58	6.13	8.39	5.38	10.05	0.43
RSPv5	Retrosplenial area ventral part layer 5	223	19	16	8.42	9.17	5.79	7.87	5.47	8.77	0.47
ACA6a	Anterior cingulate area dorsal part layer 6a	222	17	15	5.09	6.65	2.96	4.23	2.49	6.39	0.53
or	optic radiation	221	63	52	6.47	6.61	4.46	5.57	3.32	6.47	0.47
BST	Bed nuclei of the stria terminalis	207	15	13	8.43	9.83	4.97	8.66	4.52	8.45	0.43
PYR	Pyramus (VIII)	207	6	5	16.05	22.45	7.72	15.37	6.24	15.40	0.23
SSs5	Supplemental somatosensory area layer 5	203	9	9	5.56	5.60	3.69	4.75	2.92	5.62	0.53
AV	Anteroventral nucleus of thalamus	203	9	7	8.43	7.07	7.16	6.90	5.31	9.39	0.50
EPd	Endopiriform nucleus dorsal part	199	12	11	4.69	4.69	2.94	4.50	2.51	4.84	0.50
DN	Dentate nucleus	197	9	9	34.13	30.64	23.10	34.31	24.36	34.47	0.18
TRN	Tegmental reticular nucleus	193	10	8	12.65	10.92	10.22	11.72	8.59	15.55	0.17
ml	medial lemniscus	193	25	23	10.42	9.78	7.44	9.02	5.40	10.70	0.17
DEC	Declive (VI)	190	7	4	13.90	21.62	6.52	14.65	6.89	16.12	0.27
SSp-m6a	Primary somatosensory area mouth layer 6a	188	9	7	9.11	8.09	6.56	7.98	3.91	9.00	0.50
NLL	Nucleus of the lateral lemniscus	184	8	8	9.14	12.25	5.24	8.80	4.04	9.51	0.20
MEA	Medial amygdalar nucleus	184	12	12	4.70	6.20	2.76	4.32	2.54	4.62	0.40
ILA6a	Infralimbic area layer 6a	178	5	5	3.95	4.77	2.31	3.18	1.62	4.09	0.57
fp	corpus callosum posterior forceps	172	44	38	6.63	7.11	4.57	5.39	3.47	6.99	0.40
LAV	Lateral vestibular nucleus	169	11	10	24.24	28.86	12.66	25.67	12.43	23.23	0.17
AUDv5	Ventral auditory area layer 5	168	4	2	4.40	4.84	2.96	3.84	2.28	4.51	0.53
HY	Hypothalamus	168	27	20	16.55	21.45	7.45	16.59	7.37	17.10	0.30
ll	lateral lemniscus	168	20	16	10.25	13.34	5.96	9.15	4.46	10.27	0.20
V	Motor nucleus of trigeminal	166	9	7	13.53	14.71	8.55	12.61	4.94	13.48	0.20
POL	Posterior limiting nucleus of the thalamus	166	17	17	7.72	11.32	3.70	6.96	2.67	8.02	0.23
LHA	Lateral hypothalamic area	166	14	11	11.74	15.25	5.30	11.90	4.57	12.90	0.30
NOD	Nodulus (X)	165	9	6	28.86	30.23	20.10	28.81	17.32	28.24	0.23
LSv	Lateral septal nucleus ventral part	164	15	12	8.82	13.66	4.08	8.82	4.39	8.65	0.43
RSPagl6a	Retrosplenial area lateral agranular part layer 6a	161	18	13	4.99	4.92	3.41	4.26	3.09	5.48	0.52
SSp-bfd6a	Primary somatosensory area barrel field layer 6a	153	19	14	6.10	6.27	3.86	5.39	2.88	7.04	0.47
MOS2/3	Secondary motor area layer 2/3	151	17	14	3.81	5.37	1.90	3.31	1.58	3.89	0.53
PF	Parafascicular nucleus	150	8	7	11.36	15.63	5.12	11.59	4.88	13.65	0.33
RSPd5	Retrosplenial area dorsal part layer 5	150	19	15	6.95	6.89	5.54	6.26	4.71	7.59	0.47
ENTm5	Entorhinal area medial part dorsal zone layer 5	149	9	9	6.91	8.47	3.68	5.84	3.64	6.81	0.50
SNr	Substantia nigra reticular part	149	19	17	33.98	27.05	30.93	30.76	24.64	37.65	0.20
icp	inferior cerebellar peduncle	148	7	6	14.73	17.91	7.78	13.83	4.94	13.56	0.20
RSPv6a	Retrosplenial area ventral part layer 6a	147	18	17	7.79	10.36	4.36	6.86	3.38	8.39	0.40

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SSp-bfd5	Primary somatosensory area barrel field layer 5	146	18	14	7.09	7.12	5.21	6.79	4.96	7.78	0.47
ENTl6a	Entorhinal area lateral part layer 6a	145	11	8	3.14	2.80	2.27	2.98	1.71	3.05	0.60
MGv	Medial geniculate complex ventral part	143	10	10	8.42	11.57	5.29	7.69	3.84	8.56	0.43
LGv	Ventral part of the lateral geniculate complex	143	8	6	9.77	9.39	6.17	8.95	5.59	9.41	0.23
VISam5	Anteromedial visual area layer 5	142	16	13	5.12	5.49	3.63	4.00	2.91	5.23	0.53
ccb	corpus callosum body	141	43	32	6.50	8.29	3.65	5.99	2.79	7.03	0.40
VCO	Ventral cochlear nucleus	141	5	4	20.46	18.83	15.52	18.79	12.13	21.02	0.20
SMT	Submedial nucleus of the thalamus	140	6	5	9.56	7.53	7.99	8.49	7.41	11.45	0.50
PAL	Pallidum	140	15	11	19.98	24.02	9.32	19.32	7.34	20.42	0.23
MOB	Main olfactory bulb	140	3	2	16.47	9.98	15.98	16.67	16.19	16.33	0.30
RSPd6a	Retrosplenial area dorsal part layer 6a	139	21	18	7.56	8.51	4.88	6.63	3.12	7.95	0.47
cing	cingulum bundle	137	36	30	5.61	8.85	2.95	4.65	2.52	5.48	0.45
VISpm5	posteromedial visual area layer 5	136	8	8	6.13	5.95	4.23	5.50	3.22	6.47	0.50
ORBm6a	Orbital area medial part layer 6a	135	3	3	4.33	4.87	2.83	3.80	2.24	4.94	0.57
PFL	Paraflocculus	133	6	5	16.59	29.09	6.29	13.98	5.01	17.86	0.33
BMAp	Basomedial amygdalar nucleus posterior part	132	10	8	5.12	5.93	3.20	5.24	3.38	5.06	0.53
SSs6a	Supplemental somatosensory area layer 6a	132	7	7	3.42	4.06	1.75	2.46	0.82	3.09	0.57
SGN	Suprageniculate nucleus	131	10	10	6.59	5.86	5.42	5.20	3.71	6.52	0.47
ENTl3	Entorhinal area lateral part layer 3	129	13	12	3.85	4.66	2.04	3.64	1.95	4.26	0.57
VISp4	Primary visual area layer 4	129	25	21	5.59	7.60	3.22	4.39	2.94	4.99	0.50
SPVO	Spinal nucleus of the trigeminal oral part	129	8	8	19.60	26.80	8.95	19.15	6.54	19.76	0.17
ORBI5	Orbital area lateral part layer 5	128	5	4	4.98	4.81	3.20	4.50	2.66	6.03	0.57
EPv	Endopiriform nucleus ventral part	124	12	10	5.50	7.00	3.02	5.17	2.40	5.20	0.50
SSp-m5	Primary somatosensory area mouth layer 5	124	7	6	6.88	6.64	4.27	8.35	5.69	9.34	0.50
NTS	Nucleus of the solitary tract	124	20	13	14.24	18.35	6.09	15.20	7.24	15.56	0.20
RSPagl5	Retrosplenial area lateral agranular part layer 5	122	16	12	6.48	7.34	4.85	5.58	3.65	6.97	0.50
ICd	Inferior colliculus dorsal nucleus	121	5	5	20.91	27.39	10.27	20.40	8.77	21.06	0.25
sptV	spinal tract of the trigeminal nerve	120	13	9	14.62	19.15	6.35	14.70	4.06	17.90	0.17
ENTm2	Entorhinal area medial part dorsal zone layer 2	120	5	5	5.30	7.34	3.13	5.07	3.18	5.18	0.57
SSp-bfd4	Primary somatosensory area barrel field layer 4	119	13	12	5.84	5.74	3.41	5.58	3.09	5.99	0.53
VM	Ventral medial nucleus of the thalamus	118	15	9	9.54	9.63	5.45	8.36	4.70	11.42	0.40
PAR	Parasubiculum	117	9	7	10.06	15.15	4.32	7.71	3.17	10.37	0.33
alv	alveus	117	54	45	6.56	6.63	4.29	5.38	2.38	6.88	0.37
VISam6a	Anteromedial visual area layer 6a	116	16	14	4.73	6.04	2.65	3.59	2.14	5.45	0.50
PSV	Principal sensory nucleus of the trigeminal	115	7	7	11.25	15.38	6.63	9.56	5.00	12.28	0.17
MDRNd	Medullary reticular nucleus dorsal part	115	5	3	9.52	15.71	3.28	9.11	3.07	11.09	0.20
PRE	Presubiculum	115	7	7	7.02	8.10	3.94	5.43	2.78	7.49	0.28
SF	Septofimbrial nucleus	115	10	9	6.68	5.01	5.95	5.07	4.59	7.25	0.47
PA	Posterior amygdalar nucleus	113	12	12	4.89	5.73	2.47	5.30	3.07	4.63	0.43
TEa6a	Temporal association areas layer 6a	112	6	5	2.73	2.50	2.09	2.45	2.05	2.61	0.57
MGm	Medial geniculate complex medial part	111	10	10	10.18	7.66	8.26	9.51	7.41	11.75	0.43

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LSc	Lateral septal nucleus caudal (caudodorsal) part	111	11	9	5.27	5.87	3.92	4.20	2.60	5.35	0.52
ENTm6	Entorhinal area medial part dorsal zone layer 6	111	6	5	3.58	4.53	2.07	3.17	1.85	3.58	0.50
vVIIIIn	vestibular nerve	109	3	3	7.19	6.74	5.46	6.90	5.15	7.07	0.17
Alv5	Agranular insular area ventral part layer 5	109	4	4	3.56	4.79	2.00	3.13	1.83	4.18	0.57
SCsg	Superior colliculus superficial gray layer	109	19	16	8.38	10.81	4.11	6.99	3.07	8.51	0.27
ACAAd5	Anterior cingulate area dorsal part layer 5	109	7	5	4.31	3.25	3.26	3.95	2.48	4.69	0.57
PL5	Prelimbic area layer 5	107	6	6	4.47	6.05	2.66	3.41	1.77	4.56	0.57
RSPv2/3	Retrosplenial area ventral part layer 2/3	107	17	14	6.86	9.26	4.39	5.94	3.60	6.43	0.50
SSp-tr5	Primary somatosensory area trunk layer 5	107	11	11	6.29	6.86	4.16	4.23	2.14	6.66	0.50
MS	Medial septal nucleus	106	5	4	19.55	14.38	15.38	21.12	15.69	17.23	0.27
SSp-ul5	Primary somatosensory area upper limb layer 5	106	10	7	5.51	5.47	3.65	4.50	2.56	6.55	0.47
dhc	dorsal hippocampal commissure	105	18	14	7.77	8.28	5.13	6.15	3.76	7.92	0.30
AUDd5	Dorsal auditory area layer 5	104	3	3	5.59	5.41	3.63	3.88	2.17	6.39	0.53
SUT	Supratrigeminal nucleus	103	12	11	12.57	16.02	6.20	12.12	4.27	12.67	0.17
ORBm5	Orbital area medial part layer 5	102	6	5	6.21	8.75	3.11	4.97	2.49	6.47	0.57
ICc	Inferior colliculus central nucleus	100	7	6	24.08	20.15	20.01	23.64	15.62	22.91	0.20
FOTU	Folium-tuber vermis (VII)	100	6	4	24.02	29.26	13.39	22.95	12.71	23.52	0.20
LGd-sh	Dorsal part of the lateral geniculate complex shell	100	13	12	8.91	5.88	7.08	7.33	5.73	8.71	0.37
ViSp2/3	Primary visual area layer 2/3	100	20	19	5.47	5.67	3.43	4.63	2.55	5.33	0.50
CUN	Cuneiform nucleus	100	10	10	11.81	16.16	6.21	12.08	5.33	13.06	0.25
AUDp5	Primary auditory area layer 5	99	5	3	6.89	7.40	4.18	7.03	4.20	9.28	0.57
SSp-tr6a	Primary somatosensory area trunk layer 6a	97	9	9	5.90	6.80	4.29	4.58	3.43	6.40	0.53
ORBvl2/3	Orbital area ventrolateral part layer 2/3	96	5	5	6.09	7.03	3.14	5.30	2.69	6.13	0.57
AMd	Anteromedial nucleus dorsal part	92	8	7	7.27	6.18	6.01	6.32	4.60	7.83	0.50
NOT	Nucleus of the optic tract	92	20	16	11.71	12.34	6.32	9.80	4.27	12.14	0.23
SSs4	Supplemental somatosensory area layer 4	92	4	4	6.60	7.05	4.25	5.58	3.45	6.95	0.53
Ald6a	Agranular insular area dorsal part layer 6a	91	7	6	4.30	6.46	2.00	3.49	1.20	6.16	0.57
ec	external capsule	91	20	17	4.26	6.72	1.85	3.42	1.23	4.84	0.48
Ald5	Agranular insular area dorsal part layer 5	90	4	4	3.79	3.95	2.36	4.03	2.02	4.97	0.60
CLI	Central linear nucleus raphe	89	6	6	25.30	24.31	16.73	25.98	14.36	24.94	0.20
LPO	Lateral preoptic area	89	3	3	31.17	25.60	26.43	30.88	24.02	30.03	0.27
LGd-ip	Dorsal part of the lateral geniculate complex ipsilateral zone	87	10	9	10.11	5.99	9.63	9.07	8.00	10.51	0.40
ViSa4	Anterior area layer 4	87	20	20	5.18	4.87	3.77	4.21	2.75	5.00	0.52
scp	superior cerebelar peduncles	86	19	18	16.35	19.80	8.54	14.19	5.16	14.06	0.20
PVT	Paraventricular nucleus of the thalamus	84	8	8	3.36	2.75	2.89	3.06	2.49	3.53	0.53
PCN	Paracentral nucleus	84	10	6	6.98	5.28	5.64	6.45	4.65	8.68	0.53
AUDv6a	Ventral auditory area layer 6a	84	5	5	3.04	2.99	1.87	2.62	1.46	4.57	0.53
ccs	corpus callosum splenium	84	28	24	7.15	8.22	4.64	5.98	3.94	7.55	0.37
MARN	Magnocellular reticular nucleus	83	4	3	13.03	14.94	8.09	12.42	5.24	14.00	0.23
PAA	Piriform-amygdalar area	82	4	3	4.35	5.56	2.72	4.29	2.59	3.98	0.40
PCG	Pontine central gray	82	7	5	12.00	12.08	8.81	12.00	6.49	13.55	0.23

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VPLpc	Ventral posterolateral nucleus of the thalamus parvicellular part	80	11	11	9.40	11.10	6.11	8.19	5.14	10.45	0.37
OT	Olfactory tubercle	79	12	9	11.18	17.55	5.74	10.92	4.80	12.27	0.43
ACAv6a	Anterior cingulate area ventral part 6a	79	9	7	5.76	8.62	3.24	4.52	2.58	5.81	0.50
dscp	superior cerebellar peduncle decussation	79	5	5	13.64	13.06	9.90	12.95	7.50	15.63	0.20
MGd	Medial geniculate complex dorsal part	78	6	6	6.94	5.85	5.28	6.03	4.41	6.28	0.43
VISpl5	Posterolateral visual area layer 5	78	3	3	3.79	3.77	2.36	3.45	2.22	3.70	0.53
scwm	supra-callosal cerebral white matter	76	45	34	5.40	5.87	3.36	4.45	2.53	6.52	0.47
ILA5	Infralimbic area layer 5	76	5	4	4.64	4.77	2.93	4.15	2.42	4.51	0.53
LH	Lateral habenula	76	13	11	14.15	17.44	6.82	13.25	5.85	14.74	0.43
fa	corpus callosum anterior forceps	75	9	6	3.44	3.69	2.23	3.58	2.13	4.14	0.47
COApm	Cortical amygdalar area posterior part medial zone	74	8	8	6.39	8.50	2.55	5.47	2.78	6.70	0.43
chpl	choroid plexus	73	12	11	7.85	13.39	3.48	6.43	2.66	8.44	0.48
BLAp	Basolateral amygdalar nucleus posterior part	73	6	6	4.15	5.13	2.76	4.16	2.49	4.36	0.50
TRS	Triangular nucleus of septum	73	7	7	5.84	5.35	3.85	4.73	3.44	6.40	0.43
ACAv5	Anterior cingulate area ventral part layer 5	73	6	4	4.35	3.79	3.16	4.00	3.11	4.40	0.50
ORBm2/3	Orbital area medial part layer 2/3	72	4	4	4.58	5.97	2.70	4.12	2.70	3.50	0.57
VISa2/3	Anterior area layer 2/3	72	20	20	4.85	5.53	2.57	3.57	1.83	4.86	0.33
V4	fourth ventricle	71	11	11	15.74	15.07	11.58	15.22	10.22	15.10	0.17
NPC	Nucleus of the posterior commissure	71	12	11	18.13	17.96	12.62	17.00	10.16	18.38	0.23
PRP	Nucleus prepositus	70	5	4	7.71	9.21	3.40	8.52	3.39	10.47	0.23
VeCB	Vestibulocerebellar nucleus	70	6	4	19.17	21.47	12.13	16.88	7.80	19.71	0.23
PPT	Posterior pretectal nucleus	70	12	11	13.57	16.95	6.64	12.55	5.23	14.69	0.23
BLAa	Basolateral amygdalar nucleus anterior part	69	9	8	16.96	29.61	3.98	15.38	3.06	15.11	0.43
VL	lateral ventricle	68	14	11	5.23	6.95	2.75	4.19	1.86	5.18	0.37
VPMpc	Ventral posteromedial nucleus of the thalamus parvicellular part	67	10	7	11.74	13.20	6.70	11.96	6.23	12.45	0.43
fi	fimbria	66	14	12	4.58	5.27	3.09	4.21	2.40	5.08	0.50
PGRNd	Paragigantocellular reticular nucleus dorsal part	66	6	4	12.65	15.98	6.00	16.91	7.24	17.25	0.17
SCop	Superior colliculus optic layer	66	18	14	8.17	8.20	5.68	8.09	4.62	7.87	0.23
ICB	Infracerebellar nucleus	65	8	7	30.62	28.98	21.99	31.88	22.68	33.27	0.17
CM	Central medial nucleus of the thalamus	65	6	6	7.68	5.02	7.45	7.72	6.22	8.88	0.53
MDRNV	Medullary reticular nucleus ventral part	65	6	4	10.69	13.61	6.37	11.52	4.49	12.25	0.17
fiber tracts	fiber tracts	64	25	23	6.88	11.39	3.74	5.82	3.07	6.34	0.27
SPFP	Subparafascicular nucleus parvicellular part	64	15	14	10.73	14.19	6.77	9.40	4.64	12.75	0.18
LA	Lateral amygdalar nucleus	63	8	7	5.56	6.60	3.64	4.76	3.13	5.68	0.47
VII	Facial motor nucleus	63	6	5	11.70	12.40	8.06	7.79	3.42	9.60	0.27
CEAc	Central amygdalar nucleus capsular part	61	6	6	5.89	9.94	3.23	5.63	2.39	6.78	0.53
RR	Midbrain reticular nucleus retrorubral area	60	6	4	14.67	15.73	9.38	13.98	7.73	14.98	0.20
SSp-Il6a	Primary somatosensory area lower limb layer 6a	59	8	8	7.21	6.18	4.93	4.41	2.21	8.77	0.50
VISl6a	Lateral visual area layer 6a	59	6	6	6.32	7.59	3.88	4.42	2.16	4.96	0.53
HPF	Hippocampal formation	59	13	12	6.99	8.46	3.70	5.57	1.80	6.87	0.47

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HATA	Hippocampo-amygdalar transition area	59	6	6	7.51	13.79	3.48	8.01	3.12	7.12	0.40
ar	auditory radiation	58	11	11	7.31	5.56	7.04	6.89	5.45	8.78	0.43
CEAI	Central amygdalar nucleus lateral part	58	4	3	2.97	3.63	1.44	2.61	1.61	2.80	0.47
FF	Fields of Forel	56	10	10	12.51	14.85	7.00	9.74	4.94	15.98	0.17
SCdw	Superior colliculus motor related deep white layer	55	15	13	14.24	15.63	6.89	13.26	5.11	16.44	0.23
SSp-Il5	Primary somatosensory area lower limb layer 5	54	5	5	9.53	10.59	6.35	5.65	3.59	8.98	0.50
ENTl2	Entorhinal area lateral part layer 2	54	7	7	6.52	6.16	3.26	6.52	3.48	7.32	0.50
PGRNI	Paragigantocellular reticular nucleus lateral part	54	6	3	11.78	14.02	6.72	10.28	4.15	15.21	0.17
GU5	Gustatory areas layer 5	53	3	2	3.66	2.37	3.21	4.62	4.47	4.81	0.53
VISI5	Lateral visual area layer 5	52	8	8	7.58	12.09	3.67	7.05	2.73	7.73	0.60
TEa5	Temporal association areas layer 5	52	7	5	3.45	3.77	2.63	3.25	2.24	3.45	0.57
SAG	Nucleus sagulum	51	4	4	4.87	4.95	3.23	4.81	2.28	4.49	0.20
LDT	Laterodorsal tegmental nucleus	51	2	2	4.55	5.30	2.83	4.49	2.39	5.52	0.33
NDB	Diagonal band nucleus	50	5	5	11.58	15.68	5.76	12.51	4.61	12.60	0.37
VISal5	Anterolateral visual area layer 5	49	2	2	6.39	5.83	4.67	5.70	4.31	6.42	0.53
CLA	Clastrum	49	4	4	2.79	3.86	1.38	1.67	0.94	3.21	0.55
em	external medullary lamina of the thalamus	49	6	6	16.64	10.00	15.38	14.18	10.37	18.03	0.27
VISpm6a	posteromedial visual area layer 6a	49	7	7	6.14	5.94	5.07	5.23	3.22	6.71	0.50
VISpor5	Postrhinal area layer 5	48	3	2	4.31	3.43	3.27	3.97	2.78	3.94	0.62
MA	Magnocellular nucleus	48	6	6	18.11	19.64	10.83	17.71	10.14	17.31	0.30
SSp-tr4	Primary somatosensory area trunk layer 4	48	9	9	5.01	5.39	3.29	3.68	1.68	4.51	0.43
tspc	crossed tectospinal pathway	48	10	8	14.66	21.01	9.13	13.48	5.17	17.51	0.20
CTXsp	Cortical subplate	47	16	14	5.16	5.58	3.14	4.52	2.90	5.07	0.47
COApI	Cortical amygdalar area posterior part lateral zone	47	8	6	4.52	6.51	2.11	4.78	2.07	4.05	0.47
RSPv1	Retrosplenial area ventral part layer 1	47	16	12	7.52	10.67	4.28	6.61	2.91	8.91	0.37
sctv	ventral spinocerebellar tract	46	8	7	10.55	10.02	7.50	10.87	7.59	11.43	0.20
P5	Peritrigeminal zone	46	5	4	8.60	19.77	3.17	9.21	1.79	8.15	0.18
SSp-ul6a	Primary somatosensory area upper limb layer 6a	46	8	7	7.80	8.66	4.70	5.75	3.07	8.67	0.40
FN	Fastigial nucleus	46	5	3	45.44	35.42	37.82	47.66	42.66	42.32	0.18
x	Nucleus x	45	4	3	12.23	21.75	6.93	12.58	6.04	12.91	0.17
GPI	Globus pallidus internal segment	45	5	5	17.48	20.43	10.39	17.20	7.12	21.59	0.23
LING	Lingula (I)	45	5	4	17.45	22.19	10.82	19.21	11.33	20.39	0.17
SSp-bfd2/3	Primary somatosensory area barrel field layer 2/3	44	11	10	8.76	10.69	3.72	7.37	2.82	8.48	0.27
aco	anterior commissure olfactory limb	44	8	7	10.32	10.05	6.34	10.30	6.41	9.15	0.30
cpd	cerebal peduncle	44	7	7	6.35	6.52	4.22	5.79	4.22	8.71	0.23
ND	Nucleus of Darkschewitsch	44	8	8	25.09	27.53	13.79	24.21	12.82	26.98	0.25
Alp6a	Agranular insular area posterior part layer 6a	44	4	4	6.83	7.95	3.27	5.86	2.88	9.01	0.50
mlf	medial longitudinal fascicle	43	9	9	16.72	21.51	5.05	15.59	4.74	15.78	0.23
rust	rubrospinal tract	43	11	6	13.82	17.15	5.20	12.98	4.99	14.41	0.23
BMAa	Basomedial amygdalar nucleus anterior part	43	3	3	7.13	8.52	3.69	6.60	3.12	6.41	0.43
ECT5	Ectorhinal area/Layer 5	43	5	4	4.55	3.73	3.28	4.29	2.81	4.27	0.60

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ORBI2/3	Orbital area lateral part layer 2/3	43	6	4	5.71	8.12	3.11	4.78	2.58	7.12	0.52
DR	Dorsal nucleus raphe	42	2	2	7.83	12.15	3.23	7.49	2.84	7.51	0.40
SPVC	Spinal nucleus of the trigeminal caudal part	42	5	5	8.83	12.83	3.83	7.96	3.92	11.05	0.20
RPF	Retroparafascicular nucleus	42	4	3	11.96	10.68	10.64	10.73	6.81	13.28	0.23
I5	Intertrigeminal nucleus	42	4	4	22.84	15.89	21.87	24.28	18.84	20.90	0.17
ILA1	Infralimbic area layer 1	42	4	4	2.64	2.79	1.26	2.06	0.94	2.77	0.57
opt	optic tract	41	7	7	14.81	16.83	8.53	14.34	6.29	14.83	0.27
mcp	middle cerebellar peduncle	40	4	4	9.12	6.83	7.20	5.77	4.09	10.10	0.18
CEAm	Central amygdalar nucleus medial part	40	8	8	14.92	18.42	6.99	15.34	7.02	16.34	0.37
MPO	Medial preoptic area	40	4	3	9.88	13.12	5.16	10.11	4.64	9.72	0.37
VISam2/3	Anteromedial visual area layer 2/3	40	9	8	4.91	5.26	3.70	4.27	2.64	4.98	0.30
AUDp6a	Primary auditory area layer 6a	40	2	2	5.46	5.26	3.23	4.16	2.76	6.78	0.55
FL	Flocculus	39	4	4	21.94	24.31	11.71	23.19	10.52	21.53	0.27
VISpor6a	Postrhinal area layer 6a	39	2	1	3.49	3.44	2.34	2.88	1.62	3.38	0.57
CA2	Field CA2	39	18	16	10.42	10.07	6.33	9.81	5.38	11.63	0.33
OP	Olivary pretectal nucleus	39	5	5	14.09	14.64	7.56	13.27	5.83	14.02	0.27
VISam4	Anteromedial visual area layer 4	39	9	9	4.45	5.08	2.28	4.16	2.07	4.30	0.47
MH	Medial habenula	38	8	6	6.35	8.49	5.19	6.08	3.60	6.48	0.50
CL	Central lateral nucleus of the thalamus	38	9	7	8.48	6.31	7.36	7.59	6.65	10.98	0.50
cbc	cerebellar commissure	37	3	3	31.93	36.78	15.96	32.90	15.44	32.70	0.27
TR	Postpiriform transition area	37	5	3	4.09	7.01	2.15	4.77	2.12	5.00	0.57
SSp-tr2/3	Primary somatosensory area trunk layer 2/3	36	4	4	7.83	11.05	3.51	8.44	2.58	6.51	0.47
AMv	Anteromedial nucleus ventral part	36	5	4	7.33	6.10	6.21	5.79	4.27	8.81	0.50
VISli6a	Laterointermediate area layer 6a	36	4	4	4.23	4.17	3.25	2.62	2.41	4.24	0.60
SSp-n6a	Primary somatosensory area nose layer 6a	36	6	5	8.90	7.06	6.86	8.66	6.78	9.11	0.47
SNc	Substantia nigra compact part	36	7	7	27.89	20.60	23.84	28.79	27.40	29.07	0.23
ccg	genu of corpus callosum	35	14	9	3.56	4.41	1.69	3.06	1.63	4.32	0.37
PS	Parastrial nucleus	35	2	2	20.12	17.47	13.32	23.99	17.82	22.97	0.40
PC5	Parvocellular motor 5 nucleus	35	4	3	9.77	8.82	6.04	10.46	5.53	9.98	0.17
VIsp6b	Primary visual area layer 6b	33	17	15	5.57	5.46	3.83	3.92	1.90	6.00	0.37
VISC5	Visceral area layer 5	33	3	3	3.35	3.28	2.49	2.88	1.27	3.04	0.53
FS	Fundus of striatum	32	4	4	13.27	16.96	7.16	13.08	6.76	13.90	0.35
SSp-m6b	Primary somatosensory area mouth layer 6b	32	5	4	5.31	5.68	3.03	3.83	0.80	5.25	0.50
LRNm	Lateral reticular nucleus magnocellular part	31	5	3	7.55	8.97	4.54	6.47	2.95	9.01	0.17
FRP6a	Frontal pole layer 6a	30	6	6	2.93	3.65	1.32	2.17	1.00	3.53	0.53
FRP2/3	Frontal pole layer 2/3	30	3	2	4.04	4.05	2.52	3.65	2.38	5.20	0.48
ENTm1	Entorhinal area medial part dorsal zone layer 1	29	5	5	4.87	6.17	3.57	4.68	3.53	4.86	0.50
V3	third ventricle	29	11	10	8.76	9.44	6.64	7.41	3.95	7.51	0.50
FRP5	Frontal pole layer 5	28	3	2	5.15	4.26	3.96	4.24	3.46	7.15	0.57
VISa6b	Anterior area layer 6b	28	17	17	4.01	4.66	2.27	3.69	2.41	4.06	0.53
AUDp4	Primary auditory area layer 4	28	1	1	3.65	4.67	1.57	3.65	1.63	3.43	0.60

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KF	Koelliker-Fuse subnucleus	28	5	5	29.73	37.74	18.14	25.31	12.43	25.84	0.20
PR	Perireunensis nucleus	28	5	3	9.46	8.09	8.01	9.05	8.15	10.37	0.43
PeF	Perifornical nucleus	28	3	3	9.41	19.23	3.88	9.42	2.94	9.12	0.30
bsc	brachium of the superior colliculus	28	14	13	15.22	16.04	9.11	15.30	7.18	17.17	0.27
Alp2/3	Agranular insular area posterior part layer 2/3	28	2	2	2.37	2.43	1.75	2.07	1.53	2.16	0.57
amc	amygdalar capsule	27	3	2	2.76	3.67	1.44	2.49	1.44	2.48	0.53
ACAd2/3	Anterior cingulate area dorsal part layer 2/3	27	1	1	2.95	3.61	1.66	2.66	1.43	2.74	0.60
VISam6b	Anteromedial visual area layer 6b	27	12	12	4.74	7.31	2.70	4.83	1.08	5.90	0.43
VISC6a	Visceral area layer 6a	27	1	1	5.44	6.24	3.62	5.32	3.80	5.70	0.53
SSp-ul4	Primary somatosensory area upper limb layer 4	27	7	5	5.83	6.01	3.51	4.45	2.30	10.23	0.53
uf	uncinate fascicle	27	7	6	26.30	33.47	10.09	26.92	11.33	27.02	0.17
ACA6b	Anterior cingulate area ventral part 6b	26	4	4	5.35	6.89	3.32	5.91	2.60	6.42	0.57
VISrl5	Rostrolateral area layer 5	26	3	3	3.77	3.41	2.77	2.75	1.10	4.83	0.47
SSp-un5	Primary somatosensory area unassigned layer 5	26	2	2	8.77	7.44	7.27	8.10	6.07	10.37	0.50
sm	stria medullaris	26	1	1	3.63	3.04	2.76	2.62	1.35	3.11	0.50
AAA	Anterior amygdalar area	25	3	2	21.10	19.19	16.29	21.06	15.00	20.30	0.33
SPA	Subparafascicular area	25	4	4	3.69	3.93	1.75	3.42	1.74	4.54	0.40
Alv6a	Agranular insular area ventral part layer 6a	25	3	3	4.30	3.98	2.61	3.73	2.17	3.50	0.50
nst	nigrostriatal tract	24	5	5	7.75	9.41	4.56	6.14	3.16	6.22	0.23
st	stria terminalis	24	6	6	7.31	5.96	5.28	6.63	4.34	7.42	0.40
PH	Posterior hypothalamic nucleus	24	7	4	11.84	25.42	2.06	10.60	2.00	11.02	0.35
CU	Cuneate nucleus	24	3	3	4.15	3.31	3.07	3.15	1.56	3.23	0.30
INC	Interstitial nucleus of Cajal	24	6	5	21.15	19.51	14.68	22.10	20.89	26.54	0.20
NI	Nucleus incertus	24	5	4	9.54	10.57	5.31	8.89	4.89	10.18	0.27
IAD	Interanterodorsal nucleus of the thalamus	23	2	2	6.92	3.86	6.88	5.67	5.31	7.10	0.50
AT	Anterior tegmental nucleus	23	2	2	19.60	19.56	11.02	15.89	7.20	15.13	0.18
VISli5	Laterointermediate area layer 5	23	5	4	6.48	4.61	5.31	5.24	4.33	5.92	0.57
MOp6b	Primary motor area Layer 6b	23	14	13	6.74	5.91	4.09	6.95	3.40	9.46	0.50
FC	Fasciola cinerea	23	4	3	7.95	5.74	5.64	5.78	5.10	10.26	0.23
NB	Nucleus of the brachium of the inferior colliculus	23	4	4	11.73	11.26	8.12	11.25	7.09	10.71	0.27
VISpm2/3	posteromedial visual area layer 2/3	23	4	4	4.43	5.92	2.59	4.42	2.62	4.59	0.37
PL2/3	Prelimbic area layer 2/3	22	3	3	2.94	3.86	1.14	2.88	0.86	3.41	0.57
V4r	lateral recess	22	9	8	17.69	24.36	7.14	17.23	5.24	16.80	0.18
PERI6a	Perirhinal area layer 6a	22	2	2	2.72	1.50	2.47	2.42	1.93	2.75	0.60
fr	fasciculus retroflexus	22	5	4	3.75	5.95	2.17	3.74	2.30	5.73	0.35
ECU	External cuneate nucleus	21	4	2	9.59	10.59	4.76	8.41	5.09	12.00	0.17
VISI1	Lateral visual area layer 1	21	1	1	3.20	2.36	2.24	4.26	2.73	5.28	0.48
SSp-un6a	Primary somatosensory area unassigned layer 6a	21	5	4	6.61	5.76	6.19	5.16	2.96	8.04	0.30
DTN	Dorsal tegmental nucleus	21	2	2	15.55	15.09	9.06	14.12	6.65	14.63	0.23
PRC	Precommissural nucleus	21	8	7	7.89	10.05	4.15	6.60	3.10	9.57	0.33
SSp-n5	Primary somatosensory area nose layer 5	20	2	2	7.76	5.60	7.16	6.18	5.89	7.33	0.50

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SOCI	Superior olivary complex lateral part	20	5	3	15.41	16.74	8.42	14.77	7.62	28.24	0.17
pc	posterior commissure	20	6	5	23.72	30.69	11.77	21.27	12.24	25.19	0.22
ECT6a	Ectorhinal area/Layer 6a	19	4	4	3.80	3.99	2.33	3.33	2.35	3.40	0.60
VISrl4	Rostrolateral area layer 4	18	4	4	6.79	4.37	7.08	6.12	5.46	8.04	0.33
ECT2/3	Ectorhinal area/Layer 2/3	18	2	2	4.55	4.57	2.27	4.91	2.20	4.63	0.57
ILA2/3	Infralimbic area layer 2/3	18	4	4	3.35	2.43	3.18	3.29	3.28	3.63	0.57
IF	Interfascicular nucleus raphe	18	1	1	50.14	22.96	56.16	47.90	54.90	49.07	0.20
IA	Intercalated amygdalar nucleus	18	2	2	6.66	6.34	3.95	6.06	3.79	4.70	0.50
VISli4	Laterointermediate area layer 4	18	1	1	2.80	1.13	2.45	2.63	2.24	2.28	0.60
Alp5	Agranular insular area posterior part layer 5	18	3	3	3.73	2.89	3.13	3.32	2.40	3.30	0.57
RSPd2/3	Retrosplenial area dorsal part layer 2/3	18	5	4	7.65	6.85	6.03	7.24	4.77	7.24	0.47
VTA	Ventral tegmental area	17	7	7	15.17	15.89	12.85	13.99	9.09	16.05	0.30
VISpm4	posteromedial visual area layer 4	17	3	3	4.15	3.25	3.07	3.41	2.04	3.91	0.45
COAa	Cortical amygdalar area anterior part	17	2	2	5.74	7.96	2.47	5.27	2.12	5.01	0.40
ACVII	Accessory facial motor nucleus	17	3	3	13.91	16.48	7.89	14.59	10.56	13.26	0.30
DMH	Dorsomedial nucleus of the hypothalamus	16	2	2	5.56	4.91	3.45	5.03	2.87	6.20	0.38
GU6a	Gustatory areas layer 6a	16	2	1	7.61	8.50	4.78	4.49	3.53	10.41	0.47
ORBI6a	Orbital area lateral part layer 6a	16	2	2	9.31	11.00	5.29	8.14	2.69	10.09	0.57
RSPagl2/3	Retrosplenial area lateral agranular part layer 2/3	16	5	4	4.22	5.15	2.33	4.96	1.50	6.06	0.53
IGL	Intergeniculate leaflet of the lateral geniculate complex	16	3	3	7.94	8.27	4.70	8.14	4.52	8.65	0.27
SSp-bfd6b	Primary somatosensory area barrel field layer 6b	16	7	5	5.17	4.61	4.16	5.91	3.44	6.53	0.57
APr	Area prostriata	16	5	5	3.12	4.17	1.52	2.31	0.57	3.21	0.20
VISal6a	Anterolateral visual area layer 6a	15	2	2	3.58	5.89	1.98	2.25	1.05	3.33	0.23
Alv2/3	Agranular insular area ventral part layer 2/3	15	3	3	3.33	3.33	2.18	3.46	3.06	3.64	0.57
AD	Anterodorsal nucleus	15	3	1	17.25	21.43	5.77	17.69	5.86	17.30	0.33
SSp-ul2/3	Primary somatosensory area upper limb layer 2/3	15	3	2	10.23	6.13	8.91	7.52	7.21	15.20	0.40
MOs1	Secondary motor area layer 1	15	5	5	4.01	6.65	1.01	3.55	0.68	3.57	0.53
AVP	Anteroventral preoptic nucleus	15	3	2	10.77	18.30	2.87	11.29	2.92	11.26	0.40
das	dorsal acoustic stria	15	5	5	21.97	22.98	8.44	22.38	7.70	23.70	0.18
py	pyramid	14	1	1	9.39	6.83	9.31	7.15	6.93	10.35	0.20
AUDpo6b	Posterior auditory area layer 6b	14	2	2	6.19	7.60	3.19	5.17	1.85	5.76	0.50
PBG	Parabigeminal nucleus	14	3	3	14.33	13.25	11.04	12.44	6.07	18.10	0.17
SSp-tr6b	Primary somatosensory area trunk layer 6b	14	6	6	8.91	6.72	7.57	7.74	6.38	9.32	0.53
AUDd6a	Dorsal auditory area layer 6a	14	3	3	6.57	9.29	2.99	4.99	1.93	11.10	0.47
Su3	Supraoculomotor periaqueductal gray	14	4	4	4.86	5.55	3.11	4.01	2.95	5.51	0.27
SSp-un4	Primary somatosensory area unassigned layer 4	14	2	2	3.79	3.73	3.00	1.11	0.74	3.15	0.42
hbc	habenular commissure	13	2	1	5.00	4.37	3.39	4.08	2.77	5.99	0.23
AUDd2/3	Dorsal auditory area layer 2/3	13	2	2	3.85	5.97	1.39	2.99	0.78	2.81	0.60
mtg	mammillotegmental tract	13	5	5	20.61	21.11	16.63	19.10	10.43	20.80	0.22
PP	Peripeduncular nucleus	13	4	4	26.48	28.75	11.79	24.93	14.25	31.87	0.23

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POR	Superior olivary complex periolivary region	13	3	3	7.59	5.96	5.02	4.26	1.61	4.57	0.15
ACAv2/3	Anterior cingulate area ventral part layer 2/3	13	3	3	2.70	2.26	1.66	2.76	1.76	2.44	0.50
SEZ	subependymal zone	12	3	3	3.16	2.06	2.69	3.32	2.64	2.92	0.47
PL6b	Prelimbic area layer 6b	12	3	3	1.63	1.70	0.67	1.45	0.37	1.38	0.47
RL	Rostral linear nucleus raphe	12	1	1	16.13	11.63	14.24	13.56	11.99	18.24	0.32
XII	Hypoglossal nucleus	12	2	2	8.02	7.09	5.57	5.39	3.95	8.92	0.25
ORBI1	Orbital area lateral part layer 1	12	4	3	5.30	5.23	3.54	4.97	2.71	12.25	0.27
PDTg	Posterodorsal tegmental nucleus	12	2	2	26.94	28.14	13.15	27.07	10.16	26.73	0.20
PERI5	Perirhinal area layer 5	12	3	3	3.38	3.91	1.73	3.49	1.47	3.43	0.60
VISpm1	posteromedial visual area layer 1	12	2	2	1.97	0.89	1.69	1.71	1.57	1.89	0.53
VISrl6a	Rostrolateral area layer 6a	12	2	2	4.67	5.33	2.32	4.25	2.65	5.04	0.43
PIL	Posterior intralaminar thalamic nucleus	11	6	6	12.35	13.55	6.88	10.81	4.31	14.61	0.23
ACA6b	Anterior cingulate area dorsal part layer 6b	11	2	2	4.06	4.89	2.27	3.89	1.36	3.55	0.55
VTN	Ventral tegmental nucleus	11	2	2	29.47	29.88	26.71	26.70	14.31	24.48	0.22
VISpl6a	Posterolateral visual area layer 6a	11	3	3	6.70	14.68	2.31	5.69	1.73	8.37	0.45
VISrl2/3	Rostrolateral area layer 2/3	11	2	2	8.53	12.09	2.96	6.41	1.79	9.60	0.23
LIN	Linear nucleus of the medulla	11	1	1	11.72	9.56	9.90	9.24	6.75	14.39	0.22
MOp1	Primary motor area Layer 1	11	5	5	2.77	3.94	0.96	1.35	0.47	1.47	0.47
VISal4	Anterolateral visual area layer 4	11	2	2	2.56	2.53	1.27	2.12	1.29	2.21	0.57
AUDpo6a	Posterior auditory area layer 6a	10	3	3	4.14	5.47	1.60	3.40	1.57	3.44	0.47
AUDpo5	Posterior auditory area layer 5	10	2	2	6.31	7.19	3.78	5.46	2.58	6.37	0.53
VISal2/3	Anterolateral visual area layer 2/3	10	2	2	4.49	2.70	4.42	3.83	3.06	3.56	0.23
SCzo	Superior colliculus zonal layer	10	6	5	10.91	9.48	6.93	9.43	6.81	12.88	0.27
AHN	Anterior hypothalamic nucleus	10	3	2	5.86	4.61	4.15	6.39	5.13	6.08	0.37
SSp-bfd1	Primary somatosensory area barrel field layer 1	10	5	4	6.36	8.00	3.48	7.08	1.78	6.33	0.43
csc	superior colliculus commissure	10	1	1	7.74	12.41	3.43	7.95	5.00	8.40	0.33
AUDpo2/3	Posterior auditory area layer 2/3	10	1	1	2.85	3.09	1.85	3.31	1.51	2.76	0.60
SUM	Supramammillary nucleus	10	4	4	18.96	26.73	3.98	20.39	1.78	18.39	0.27
SSp-n4	Primary somatosensory area nose layer 4	10	1	1	4.73	3.47	4.24	2.22	1.78	3.66	0.20
VISam1	Anteromedial visual area layer 1	10	4	4	3.41	6.18	0.59	3.47	0.68	3.81	0.40
VISp1	Primary visual area layer 1	9	6	6	4.27	2.97	4.40	3.11	3.40	3.96	0.27
cst	corticospinal tract	9	1	1	9.38	6.39	8.70	8.59	6.12	12.50	0.20
IG	Induseum griseum	9	3	3	3.28	2.52	2.94	3.12	1.55	5.16	0.57
bic	brachium of the inferior colliculus	9	4	4	7.50	7.44	5.14	8.08	5.07	8.25	0.30
GR	Gracile nucleus	9	1	1	12.47	9.05	10.21	12.07	9.79	13.52	0.23
DT	Dorsal terminal nucleus of the accessory optic tract	9	2	2	6.26	2.66	6.35	5.64	5.18	6.03	0.27
mtt	mammillothalamic tract	9	2	1	9.18	8.42	5.01	6.12	4.71	8.85	0.23
PG	Pontine gray	9	3	3	12.07	10.13	9.02	10.04	5.55	11.92	0.17
SSs6b	Supplemental somatosensory area layer 6b	9	4	4	3.33	3.03	1.95	3.12	1.57	3.74	0.65
TU	Tuberal nucleus	8	1	1	6.59	7.61	3.35	6.49	3.83	6.90	0.27
LM	Lateral mammillary nucleus	8	1	1	8.42	7.48	5.43	6.91	4.21	12.32	0.27

Acronym	Region (NWB location)	Units	Sessions	Mice	firing_rate mean (Hz)	SD (Hz)	Median (Hz)	Pre-stim mean (Hz)	Pre-stim median (Hz)	Stim mean (Hz)	peak_to_tro ugh median (ms)
AMBv	Nucleus ambiguus ventral division	8	2	1	3.80	2.91	3.12	4.38	3.23	5.13	0.17
RE	Nucleus of reuniens	8	3	3	7.52	9.76	4.46	6.49	3.94	6.90	0.50
BLAv	Basolateral amygdalar nucleus ventral part	8	2	2	3.34	3.27	2.51	3.70	2.74	3.08	0.37
SSp-m4	Primary somatosensory area mouth layer 4	8	3	3	4.99	3.53	4.39	6.89	5.30	7.68	0.50
SSp-lI4	Primary somatosensory area lower limb layer 4	8	3	3	7.75	10.49	3.16	1.77	1.83	3.57	0.33
PSTN	Parasubthalamic nucleus	8	2	2	15.60	11.79	18.20	15.07	14.57	15.47	0.20
tb	trapezoid body	7	1	1	6.95	6.35	5.26	6.66	2.83	5.84	0.23
AId6b	Agranular insular area dorsal part layer 6b	7	2	2	6.86	11.01	2.94	6.03	2.51	9.02	0.35
IAM	Interanteromedial nucleus of the thalamus	7	1	1	3.57	1.04	3.40	2.79	2.53	3.77	0.60
SubG	Subgeniculate nucleus	7	1	1	9.23	6.30	11.85	11.04	13.07	11.09	0.30
RSPag1	Retrosplenial area lateral agranular part layer 1	7	3	3	4.70	3.07	5.66	5.12	4.88	4.97	0.43
III	Oculomotor nucleus	7	3	3	5.48	4.83	3.19	7.33	5.94	7.22	0.27
VISl6b	Lateral visual area layer 6b	7	2	2	8.64	3.42	8.68	6.73	6.96	7.93	0.52
VISal6b	Anterolateral visual area layer 6b	7	1	1	4.18	4.90	2.17	3.49	2.40	4.47	0.57
act	anterior commissure temporal limb	6	3	3	6.72	8.65	3.52	7.62	5.83	5.51	0.23
SPFm	Subparafascicular nucleus magnocellular part	6	3	3	7.99	7.58	7.23	7.37	6.22	8.20	0.57
SSp-un2/3	Primary somatosensory area unassigned layer 2/3	6	1	1	6.07	2.86	6.83	0.00	0.00	0.00	0.62
VISl2/3	Lateral visual area layer 2/3	6	3	3	11.83	17.58	3.43	16.05	2.99	16.32	0.27
VISli6b	Laterointermediate area layer 6b	6	2	2	3.01	2.75	2.11	3.35	2.72	3.82	0.47
STN	Subthalamic nucleus	6	3	3	14.39	11.20	13.63	15.40	15.00	21.15	0.17
VISl4	Lateral visual area layer 4	6	2	2	11.86	12.27	7.12	12.40	9.09	13.95	0.27
VISpor2/3	Postrhinal area layer 2/3	6	2	2	2.36	2.12	1.74	1.04	1.03	2.98	0.55
ee	corpus callosum extreme capsule	6	4	4	4.09	4.44	2.20	3.59	1.26	3.42	0.37
VISrl6b	Rostrolateral area layer 6b	6	4	4	2.81	1.65	2.32	2.60	1.35	1.94	0.60
y	Nucleus y	6	1	1	51.46	22.51	48.20	52.78	47.71	51.42	0.15
IPDM	Interpeduncular nucleus dorsomedial	6	1	1	55.83	22.40	48.22	49.62	46.06	53.86	0.27
IntG	Intermediate geniculate nucleus	6	1	1	8.57	8.13	5.30	5.00	3.69	12.37	0.17
RSPd1	Retrosplenial area dorsal part layer 1	6	3	3	10.45	10.76	5.44	9.15	1.64	10.76	0.27
PL1	Prelimbic area layer 1	6	2	2	2.31	2.84	1.40	1.85	0.72	1.25	0.52
RH	Rhomboid nucleus	6	2	2	4.28	3.04	4.21	2.48	1.24	5.91	0.50
mfb	medial forebrain bundle	6	1	1	4.64	6.98	2.49	3.86	0.93	5.94	0.23
PVHd	Paraventricular hypothalamic nucleus descending division	6	2	2	17.20	7.05	17.18	16.09	16.03	14.92	0.43
VISpl4	Posterolateral visual area layer 4	6	1	1	1.73	2.24	0.99	1.54	0.70	1.21	0.50
RSPd6b	Retrosplenial area dorsal part layer 6b	5	2	2	3.50	4.28	1.89	2.30	0.74	2.86	0.47
ADP	Anterodorsal preoptic nucleus	5	1	1	27.14	26.81	10.86	29.44	15.37	27.96	0.37
PVH	Paraventricular hypothalamic nucleus	5	2	2	8.05	2.98	6.63	6.12	7.55	10.67	0.28
ECT6b	Ectorhinal area/Layer 6b	5	1	1	1.93	1.35	2.15	2.11	1.04	2.20	0.53
VISpor6b	Postrhinal area layer 6b	5	2	2	3.13	1.99	2.07	2.44	2.67	3.07	0.52
MMI	Medial mammillary nucleus lateral part	5	1	1	5.76	9.45	0.50	6.14	0.56	7.91	0.20
SSp-lI2/3	Primary somatosensory area lower limb layer 2/3	5	3	3	5.67	5.41	3.03	2.58	1.47	4.45	0.30
MEPO	Median preoptic nucleus	5	1	1	18.09	23.74	9.45	17.88	9.88	18.09	0.23

Acronym	Region (NWB location)	Units	Sessions	Mice	firing_rate mean (Hz)	SD (Hz)	Median (Hz)	Pre-stim mean (Hz)	Pre-stim median (Hz)	Stim mean (Hz)	peak_to_tro ugh median (ms)
VISa1	Anterior area layer 1	4	2	2	11.85	13.23	10.27	10.72	9.71	7.57	0.27
VISpm6b	posteromedial visual area layer 6b	4	2	2	2.56	1.55	2.53	2.80	2.46	2.42	0.55
SSp-n6b	Primary somatosensory area nose layer 6b	4	3	1	6.55	3.34	7.98	6.58	7.36	7.25	0.38
ACAv1	Anterior cingulate area ventral part layer 1	4	2	2	1.17	0.43	1.11	0.73	0.61	1.43	0.50
SH	Septohippocampal nucleus	4	2	2	2.80	1.68	3.09	2.12	1.42	2.22	0.58
Pa5	Paratrigeminal nucleus	4	1	1	5.50	3.64	6.09	4.78	4.53	5.88	0.28
ISN	Inferior salivatory nucleus	4	1	1	65.48	38.56	50.21	73.40	59.07	63.16	0.13
RSPagl6b	Retrosplenial area lateral agranular part layer 6b	4	3	3	13.08	14.71	8.72	6.27	2.75	25.06	0.33
RSPv6b	Retrosplenial area ventral part layer 6b	4	3	3	21.03	21.05	15.76	14.27	8.95	27.95	0.32
PT	Parataenial nucleus	4	1	1	2.04	1.50	1.83	2.09	2.08	1.92	0.60
ORBm1	Orbital area medial part layer 1	4	2	2	3.13	3.61	1.49	3.27	1.54	2.00	0.45
ORBv1	Orbital area ventrolateral part layer 1	4	1	1	10.53	15.01	4.53	9.69	4.52	12.98	0.68
AP	Area postrema	3	1	1	5.16	4.52	5.54	9.19	9.22	12.39	0.25
MOs6b	Secondary motor area layer 6b	3	3	3	4.74	2.17	5.63	5.76	5.26	8.50	0.55
SSs2/3	Supplemental somatosensory area layer 2/3	3	2	1	3.42	2.08	3.12	1.74	1.73	4.20	0.27
sV	sensory root of the trigeminal nerve	3	2	2	20.13	17.14	22.61	11.57	13.47	33.87	0.20
Pa4	Paratrochlear nucleus	3	1	1	40.55	48.67	24.63	37.41	23.06	46.02	0.20
VISal1	Anterolateral visual area layer 1	3	1	1	2.06	1.08	1.77	0.78	0.50	0.63	0.53
SSp-m2/3	Primary somatosensory area mouth layer 2/3	2	1	1	0.90	0.49	0.90	0.09	0.09	0.04	
Alp6b	Agranular insular area posterior part layer 6b	2	2	2	2.58	1.71	2.58	0.44	0.44	0.99	0.23
moV	motor root of the trigeminal nerve	2	1	1	2.28	1.84	2.28	1.43	1.43	1.69	0.12
VMH	Ventromedial hypothalamic nucleus	2	1	1	3.44	1.47	3.44	4.41	4.41	3.98	
SSp-Il6b	Primary somatosensory area lower limb layer 6b	2	2	2	9.11	6.54	9.11	6.15	6.15	8.08	0.40
vhc	ventral hippocampal commissure	2	1	1	3.01	1.77	3.01	3.18	3.18	3.24	0.35
VISrl1	Rostrolateral area layer 1	2	2	2	4.15	5.24	4.15	1.24	1.24	5.22	0.23
IVn	trochlear nerve	2	1	1	5.91	5.56	5.91	6.26	6.26	9.77	0.32
AUDpo4	Posterior auditory area layer 4	2	2	2	2.68	1.66	2.68	3.25	3.25	5.49	0.45
SSp-Il1	Primary somatosensory area lower limb layer 1	2	1	1	4.41	5.06	4.41	2.04	2.04	3.88	0.27
SOCm	Superior olivary complex medial part	2	1	1	13.43	9.72	13.43	13.86	13.86	13.75	0.17
DMX	Dorsal motor nucleus of the vagus nerve	2	1	1	8.38	4.59	8.38	2.84	2.84	8.30	0.28
PMd	Dorsal premammillary nucleus	2	1	1	10.20	13.07	10.20	5.85	5.85	9.73	0.20
SSp-n2/3	Primary somatosensory area nose layer 2/3	2	1	1	2.19	1.61	2.19	1.79	1.79	1.40	0.38
AQ	cerebral aqueduct	2	2	2	17.17	7.03	17.17	19.17	19.17	26.20	0.23
AUDd6b	Dorsal auditory area layer 6b	2	1	1	12.87	14.49	12.87	6.36	6.36	11.11	0.20
ENTl1	Entorhinal area lateral part layer 1	2	2	2	0.97	0.48	0.97	0.78	0.78	0.64	0.52
IPA	Interpeduncular nucleus apical	2	1	1	36.35	48.19	36.35	31.09	31.09	36.08	
IO	Inferior olivary complex	2	1	1	7.98	7.42	7.98	8.84	8.84	9.54	
MPN	Medial preoptic nucleus	2	1	1	4.28	3.45	4.28	4.21	4.21	2.56	0.40
TEa6b	Temporal association areas layer 6b	2	1	1	5.26	4.06	5.26	5.45	5.45	5.84	0.27
PAS	Parasolitary nucleus	1	1	1	2.32		2.32	0.08	0.08	0.25	0.17
RO	Nucleus raphe obscurus	1	1	1	1.62		1.62	1.97	1.97	1.83	0.30

Acronym	Region (NWB location)	Units	Sessions	Mice	firing_rate mean (Hz)	SD (Hz)	Median (Hz)	Pre-stim mean (Hz)	Pre-stim median (Hz)	Stim mean (Hz)	peak_to_tro ugh median (ms)
pyd	pyramidal decussation	1	1	1	1.48		1.48	0.31	0.31	0.13	
Alp1	Agranular insular area posterior part layer 1	1	1	1	11.45		11.45	9.20	9.20	8.76	0.27
SSp-ul1	Primary somatosensory area upper limb layer 1	1	1	1	6.06		6.06	1.34	1.34	1.71	
SSp-ul6b	Primary somatosensory area upper limb layer 6b	1	1	1	3.13		3.13	2.88	2.88	1.79	0.17
PERI2/3	Perirhinal area layer 2/3	1	1	1	1.78		1.78	1.66	1.66	1.63	0.53
PERI1	Perirhinal area layer 1	1	1	1	3.90		3.90	4.29	4.29	4.26	0.53
VISC6b	Visceral area layer 6b	1	1	1	6.85		6.85	0.76	0.76	3.31	0.20
SSp-tr1	Primary somatosensory area trunk layer 1	1	1	1	0.29		0.29	0.39	0.39	0.43	
IMD	Intermediodorsal nucleus of the thalamus	1	1	1	2.49		2.49	3.03	3.03	1.97	0.47
IPDL	Interpeduncular nucleus dorsolateral	1	1	1	54.35		54.35	61.52	61.52	61.53	0.17
IPI	Interpeduncular nucleus intermediate	1	1	1	2.60		2.60	0.10	0.10	0.06	0.47
fx	columns of the fornix	1	1	1	3.79		3.79	5.32	5.32	4.95	0.20
AUDv6b	Ventral auditory area layer 6b	1	1	1	0.42		0.42	0.12	0.12	0.15	0.63
VISpor1	Postrhinal area layer 1	1	1	1	5.17		5.17	4.69	4.69	3.90	

6. Subjects

Subject	Lab	Sex	Sessions	Median age (d)	Trials	is_mouse_rewarded (%)	Units	Score = 1
CSHL045	churchlandlab	M	5	204	3950	81.0	6046	731
CSHL047	churchlandlab	M	2	168	1121	81.3	2818	304
CSHL049	churchlandlab	M	4	156	2880	80.8	4336	581
CSHL051	churchlandlab	F	1	176	510	77.6	1805	133
CSHL052	churchlandlab	F	5	190	4393	81.4	6195	593
CSHL053	churchlandlab	F	4	171	2560	85.2	4274	664
CSHL054	churchlandlab	F	1	182	855	75.7	1307	119
CSHL055	churchlandlab	F	1	188	771	74.1	2846	171
CSHL058	churchlandlab	M	3	267	1676	80.0	4524	569
CSHL059	churchlandlab	M	5	143	2994	84.4	7041	646
CSHL060	churchlandlab	M	5	149	3405	85.1	9229	1154
CSH_ZAD_001	zadorlab	F	1	275	854	87.4	519	50
CSH_ZAD_011	zadorlab	F	3	187	1933	85.8	5287	458
CSH_ZAD_019	zadorlab	F	4	279	3460	84.0	7351	798
CSH_ZAD_022	zadorlab	M	8	194	7464	83.0	11932	1387
CSH_ZAD_024	zadorlab	M	1	209	751	88.3	2522	380
CSH_ZAD_025	zadorlab	M	7	122	5514	83.6	14956	1676
CSH_ZAD_026	zadorlab	M	6	140	4569	85.2	11387	1726
CSH_ZAD_029	zadorlab	M	7	168	8463	89.0	16291	1916
DY_008	danlab	M	2	198	880	80.9	1699	250
DY_009	danlab	M	9	199	3907	81.5	7534	1187
DY_010	danlab	M	5	159	2588	80.2	3229	517
DY_011	danlab	M	4	170	2024	81.8	1940	267
DY_013	danlab	M	3	136	1242	80.7	2931	275
DY_014	danlab	M	7	262	4340	80.1	7757	1091
DY_016	danlab	M	5	123	2461	74.4	10248	1514
DY_018	danlab	M	3	157	1658	77.9	4406	496
DY_020	danlab	M	8	142	4714	80.5	12942	1934
KS014	cortexlab	M	3	219	1876	84.9	3919	538
KS016	cortexlab	M	3	222	1742	81.1	3472	349
KS022	cortexlab	M	1	167	826	85.0	925	89
KS023	cortexlab	F	4	166	2809	83.8	5461	823
KS042	cortexlab	M	2	138	1149	67.7	3596	362
KS043	cortexlab	M	2	152	1095	83.0	2948	257
KS044	cortexlab	M	2	153	1257	80.1	4342	340
KS045	cortexlab	M	1	151	404	75.7	1949	171
KS046	cortexlab	M	4	148	2887	76.7	6744	814
KS051	cortexlab	M	3	169	1298	77.7	1842	224
KS052	cortexlab	M	1	170	691	80.6	755	91
KS055	cortexlab	M	2	160	923	74.4	4416	598

Subject	Lab	Sex	Sessions	Median age (d)	Trials	is_mouse_rewarded (%)	Units	Score = 1
KS084	cortexlab	M	1	167	437	67.5	911	131
KS086	cortexlab	M	2	210	1103	76.9	2334	257
KS091	cortexlab	M	2	202	1492	74.9	4143	221
KS094	cortexlab	M	4	136	2165	79.9	7315	623
KS096	cortexlab	M	5	136	3460	80.6	7325	725
MFD_05	churchlandlab_ucla	M	3	169	1837	79.8	3689	481
MFD_06	churchlandlab_ucla	M	4	176	2954	81.6	5668	543
MFD_07	churchlandlab_ucla	M	3	184	2361	79.1	3321	560
MFD_08	churchlandlab_ucla	M	2	192	1529	77.8	2396	436
MFD_09	churchlandlab_ucla	M	1	233	569	70.3	1557	291
NR_0017	steinmetzlab	M	1	429	911	81.0	758	60
NR_0019	steinmetzlab	M	3	273	1959	87.6	3152	459
NR_0020	steinmetzlab	M	4	236	2573	89.0	6815	700
NR_0021	steinmetzlab	M	3	230	1653	84.5	2989	384
NR_0024	steinmetzlab	F	4	456	2426	83.2	7862	576
NR_0027	steinmetzlab	M	4	171	4119	79.4	5467	630
NR_0028	steinmetzlab	F	5	279	4324	81.9	8657	943
NR_0029	steinmetzlab	F	5	197	2191	82.2	1992	115
NR_0031	steinmetzlab	F	2	225	893	87.7	3489	390
NYU-11	angelakilab	F	2	212	990	85.4	2626	340
NYU-12	angelakilab	F	4	148	2132	82.7	6147	655
NYU-21	angelakilab	F	2	229	1393	75.4	3192	368
NYU-30	angelakilab	M	3	155	2434	88.3	2898	398
NYU-37	angelakilab	M	5	149	2455	84.2	9526	1278
NYU-39	angelakilab	M	5	155	2698	86.5	6815	694
NYU-40	angelakilab	M	4	128	1959	85.1	5704	806
NYU-45	angelakilab	M	3	168	1415	93.2	4208	701
NYU-46	angelakilab	M	4	142	1849	83.9	3402	225
NYU-47	angelakilab	M	3	142	1523	82.1	5316	480
NYU-48	angelakilab	M	3	130	1875	91.4	5206	557
NYU-65	angelakilab	M	3	135	1445	79.8	6159	715
PL015	hausserlab	M	5	283	2936	91.1	4694	396
PL016	hausserlab	M	1	264	426	79.8	1405	155
PL017	hausserlab	M	7	184	4489	85.9	7713	832
PL024	hausserlab	M	1	199	416	92.1	1035	126
PL030	hausserlab	M	8	332	3872	84.2	7214	1241
PL031	hausserlab	M	5	370	2817	84.8	2456	351
PL033	hausserlab	M	13	334	7501	83.1	7510	1104
PL034	hausserlab	M	3	116	2180	87.7	1234	124
PL035	hausserlab	M	4	286	2090	84.2	2851	267
PL037	hausserlab	M	6	282	3685	80.8	6903	932
PL050	hausserlab	M	9	115	6649	86.6	12144	1282

Subject	Lab	Sex	Sessions	Median age (d)	Trials	is_mouse_rewarded (%)	Units	Score = 1
SWC_038	mrsicflogellab	F	3	233	2257	72.1	7231	724
SWC_039	mrsicflogellab	F	1	274	403	80.6	2612	207
SWC_042	hoferlab	M	2	217	1027	73.5	3322	234
SWC_043	hoferlab	M	7	282	4518	80.7	11250	1352
SWC_052	mrsicflogellab	F	7	161	4068	80.8	12918	802
SWC_053	mrsicflogellab	M	3	183	1533	73.0	4701	256
SWC_054	mrsicflogellab	M	6	149	3915	77.3	9615	1039
SWC_058	mrsicflogellab	M	4	134	2872	73.8	8104	778
SWC_060	hoferlab	F	3	120	1732	78.4	3778	404
SWC_061	hoferlab	F	3	121	1785	78.7	4394	483
SWC_065	mrsicflogellab	F	1	102	839	79.6	1093	59
SWC_066	hoferlab	F	2	92	1348	72.8	847	22
UCLA005	churchlandlab_ucla	F	1	181	772	80.1	1108	53
UCLA006	churchlandlab_ucla	F	1	309	748	84.6	1242	314
UCLA011	churchlandlab_ucla	M	1	168	764	78.0	684	62
UCLA012	churchlandlab_ucla	M	1	155	719	77.3	1187	98
UCLA014	churchlandlab_ucla	F	1	155	581	75.9	983	99
UCLA015	churchlandlab_ucla	F	1	380	610	78.5	1186	133
UCLA017	churchlandlab_ucla	F	1	245	665	80.9	1248	257
UCLA030	churchlandlab_ucla	F	1	301	505	68.3	1316	343
UCLA033	churchlandlab_ucla	F	3	226	2143	75.1	4657	584
UCLA034	churchlandlab_ucla	M	4	182	1852	73.0	7001	1193
UCLA035	churchlandlab_ucla	M	2	208	1750	80.6	4434	452
UCLA036	churchlandlab_ucla	M	1	224	476	75.4	2681	304
UCLA037	churchlandlab_ucla	M	4	196	2194	82.3	4375	805
UCLA044	churchlandlab_ucla	F	2	181	1261	75.4	1638	201
UCLA048	churchlandlab_ucla	F	1	245	458	62.7	1242	162
UCLA049	churchlandlab_ucla	F	2	236	910	70.0	3003	375
UCLA052	churchlandlab_ucla	F	1	210	436	79.8	914	93
ZFM-01576	mainenlab	F	2	190	1855	81.0	2172	537
ZFM-01577	mainenlab	F	3	156	1899	80.8	4871	641
ZFM-01592	mainenlab	M	3	129	2726	84.3	5214	935
ZFM-01935	mainenlab	M	4	166	3150	81.0	3720	692
ZFM-01936	mainenlab	F	5	152	2570	79.8	4937	827
ZFM-01937	mainenlab	F	2	228	1233	83.9	1393	217
ZFM-02368	mainenlab	M	1	163	408	82.1	1719	372
ZFM-02369	mainenlab	M	1	150	733	81.7	1864	474
ZFM-02370	mainenlab	M	2	128	1367	78.6	1567	356
ZFM-02372	mainenlab	F	2	164	1068	81.9	1892	247
ZFM-02373	mainenlab	M	1	185	424	77.6	1547	367
ZFM-04308	mainenlab	F	2	236	817	72.7	2977	253
ZM_1897	mainenlab	F	2	177	961	86.2	3770	668

Subject	Lab	Sex	Sessions	Median age (d)	Trials	is_mouse_rewarded (%)	Units	Score = 1
ZM_1898	mainenlab	F	2	184	1398	81.0	1749	203
ZM_2240	mainenlab	M	4	164	2777	84.3	6450	1153
ZM_2241	mainenlab	F	5	170	3647	85.7	8083	1550
ZM_2245	mainenlab	F	1	162	872	85.9	1368	201
ZM_3003	mainenlab	M	2	260	1236	80.1	2084	271
ibl_witten_13	wittenlab	M	3	218	2420	89.6	4162	567
ibl_witten_17	wittenlab	M	3	253	2237	80.8	2109	153
ibl_witten_18	wittenlab	M	3	294	2373	79.6	3140	327
ibl_witten_19	wittenlab	M	3	252	1688	80.3	2948	295
ibl_witten_20	wittenlab	M	3	294	2879	83.0	5246	635
ibl_witten_25	wittenlab	M	1	182	563	81.7	2055	308
ibl_witten_26	wittenlab	M	5	234	3804	78.3	8538	900
ibl_witten_27	wittenlab	F	6	188	4118	79.4	8969	962
ibl_witten_29	wittenlab	M	9	258	6241	86.2	7834	881
ibl_witten_32	wittenlab	M	1	295	406	74.1	672	33

7. Data checks performed by this export

Region labels. 2 distinct location names (12,215 units) did not match an Allen ontology name and have an empty *region* acronym; their *region_name* is kept. 'void' (12,210), 'out-of-atlas' (5)

peak_to_trough_duration_ms. 208,811 units have the value -1 in the NWB file; they are treated as missing.

8. Column and signal descriptions from the NWB files

Quoted verbatim from sub-SWC-066_ses-a4747ac8-6a75-444f-b99b-696fff0243fd_desc-processed_behavior+ecephys.nwb; all files in the Dandiset share this structure.

Location in NWB file	Name	Description (as written in the NWB file)
electrodes	(table)	metadata about extracellular electrodes
electrodes	location	Location of the electrode (channel).
electrodes	group	Reference to the ElectrodeGroup.
electrodes	electrode_name	Unique identifier derived from probe contact ids.
electrodes	contact_shapes	Contact shape per electrode as defined by the probe.
electrodes	channel_name	The name of this channel, showing all recording streams for the electrode (e.g., 'AP379,LF379').
electrodes	rel_x	Relative x coordinate on the probe (μm).
electrodes	rel_y	Relative y coordinate on the probe (μm).
electrodes	probe_name	IBL probe name in canonical format (e.g., 'Probe00', 'Probe01').
electrodes	contact_ids	Channel IDs as specified by the probe manufacturer.
electrodes	shank_ids	Shank ID for each electrode (for multi-shank probes).
electrodes	adc_group	ADC group index from SpikeGLX metadata.
electrodes	adc_sample_order	ADC sample order from SpikeGLX metadata.
electrodes	group_name	Name of the ElectrodeGroup.
electrodes	x	x coordinate of the channel location in the brain.
electrodes	y	y coordinate of the channel location in the brain.
electrodes	z	z coordinate of the channel location in the brain.
electrodes	imp	Impedance of the channel, in ohms.
electrodes	filtering	Description of hardware filtering.
units	(table)	Spike-sorted units from Neuropixels 1.0 probes. Spike sorting was performed using a Python implementation of Kilosort 2.5 (iblsorter, formerly pykilosort) with IBL-specific optimizations for drift correction and automated quality control. Unit quality is determined by three metrics: (1) refractory period violations using the slidingRP algorithm (Hill et al. 2011), (2) noise cutoff based on amplitude distribution, and (3) median spike amplitude threshold ($>50\text{ uV}$). The 'label' column indicates the proportion of metrics passed (1.0 = all passed, i.e. 'good' unit). Each unit includes brain region assignment from IBL's histology pipeline with Allen CCF coordinates. For methodology details, see the IBL spike sorting white paper: https://doi.org/10.6084/m9.figshare.19705522
units	unit_name	Unique reference for each unit.
units	min_spike_amplitude_uV	Minimum spike amplitude in microvolts.
units	max_spike_amplitude_uV	Maximum spike amplitude in microvolts.
units	firing_rate	Average firing rate in Hz.
units	probe_name	Name of probe this unit was recorded from in canonical format (e.g., 'Probe00', 'Probe01').
units	spike_amplitude_std_dB	Standard deviation of log-transformed spike amplitudes in decibels. High values ($>6\text{ dB}$) may indicate drift or contamination.
units	isi_violations_ratio	Ratio of observed to expected ISI violations (Steinmetz/cortex-lab method). Can exceed 1.0. Equivalent to SpikeInterface's isi_violations_ratio.
units	rp_violation	Contamination estimate using Hill et al. (2011) quadratic formula. Aligns with SpikeInterface's rp_violations.
units	cluster_uuid	Globally unique identifier for cross-referencing with IBL database via ONE API.
units	presence_ratio	Fraction of time bins (60s default) with at least one spike. Good units typically >0.9 .
units	kilosort2_label	Original Kilosort2 classification: 'good', 'mua', or 'noise'.
units	peak_to_trough_duration_ms	Duration from negative trough to positive peak of the mean waveform in milliseconds. Used to distinguish fast-spiking interneurons ($\sim 0.2\text{-}0.4\text{ ms}$) from regular-spiking pyramidal cells ($\sim 0.4\text{-}0.8\text{ ms}$).
units	median_spike_amplitude_uV	Median spike amplitude in microvolts. IBL uses 50 uV threshold for quality filtering.
units	missed_spikes_estimate	Estimated fraction of spikes missed due to detection threshold (0.0-1.0).
units	spike_times	the spike times for each unit in seconds
units	electrodes	the electrodes that each spike unit came from

Location in NWB file	Name	Description (as written in the NWB file)
units	presence_ratio_std	Standard deviation of spike counts across 10-second time bins.
units	waveform_mean	the spike waveform mean for each spike unit
units	sliding_rp_violation	Binary pass/fail (0.0 or 1.0) using sliding refractory period algorithm. Primary metric for ibl_quality_score. Aligns with SpikeInterface's sliding_rp_violations.
units	ibl_quality_score	Proportion of IBL quality metrics passed (0.0, 0.33, 0.67, or 1.0). 1.0 = all three passed.
units	spike_distances_from_probe_tip_um	Distance from the probe tip for each spike in micrometers, computed from waveform center of mass. 0 = probe tip, values increase toward brain surface.
units	spike_amplitudes_uV	Peak amplitude of each spike for each unit in microvolts.
units	cumulative_drift_um_per_hour	Sum of absolute depth changes between consecutive spikes, normalized to um/hour. Formula: $\text{sum}(\text{abs}(\text{diff}(\text{spike_depths}))) / \text{duration} * 3600$. High values indicate either electrode drift or depth estimation noise. Scales with spike count (-0.79 correlation). NOT actual electrode displacement.
units	max_electrode	Index into the electrodes table for the electrode with maximum spike amplitude for this unit.
units	distance_from_probe_tip_um	Mean distance from the probe tip in micrometers, computed from waveform center of mass. 0 = probe tip, values increase toward brain surface.
units	noise_cutoff	Standard deviations the lower amplitude quartile deviates from expected. Large values indicate missed spikes at detection threshold.
units	spike_count	Total number of spikes assigned to this unit. Integer type; -1 indicates missing data.
intervals/trials	(table)	Trial data from the IBL decision-making task. On each trial, a visual stimulus (Gabor patch) appears on the left or right of a screen, and the mouse must move it to the center by turning a wheel with its front paws within 60 seconds. Correct responses are rewarded with water; incorrect responses trigger a white noise auditory cue and 2-second timeout. If no response is recorded during 60 seconds, a white noise auditory cue is presented. Trial timeline: (1) start_time begins a quiescence period where the mouse must hold the wheel still; (2) after quiescence_period elapses, the stimulus and auditory go cue are presented (gabor_stimulus_onset_time); (3) the mouse responds by turning the wheel to move the stimulus; correct choices are defined by moving the visual stimulus into the center; (4) feedback_time marks reward punishment or presentation of the white noise auditory cue; (5) stop_time marks trial end. Stimulus contrast is uniformly sampled from 5 values (0%, 6.25%, 12.5%, 25%, 100%). On 0% contrast trials, no stimulus is visible but a correct side is still assigned based on block probability, allowing mice to perform above chance using prior information. After an initial 90 unbiased trials, stimulus probability alternates between left-biased (80:20) and right-biased (20:80) blocks of 20-100 trials (mean 51). Block changes are not cued. See IBL et al. (2021) eLife 10:e63711 for full task details.
intervals/trials	start_time	The beginning of the trial.
intervals/trials	stop_time	The end of the trial.
intervals/trials	quiescence_period	Required duration (seconds) the mouse must hold the wheel still before stimulus presentation. Sampled from exponential distribution (400-700ms, mean ~550ms).
intervals/trials	gabor_stimulus_onset_time	Time when the visual stimulus (Gabor patch) appears on the screen, detected by the photodiode. Presented together with the auditory go cue.
intervals/trials	auditory_cue_time	Time of the auditory go cue (100ms, 5kHz tone) signaling the mouse may respond. Presented together with the visual stimulus.
intervals/trials	wheel_movement_onset_time	Time of first detected wheel movement (≥ 0.1 radians threshold) after the go cue.
intervals/trials	choice_registration_time	Time when the mouse's choice was registered: either the wheel movement reached the +/-35 degree threshold, or 60-second timeout elapsed.
intervals/trials	feedback_time	Time of feedback delivery: water reward for correct responses, or white noise pulse + 2-second timeout for incorrect responses.
intervals/trials	gabor_stimulus_offset_time	Time when the Gabor patch disappears from the screen, detected by the photodiode.
intervals/trials	gabor_stimulus_contrast	Contrast of the Gabor patch as a percentage (0, 6.25, 12.5, 25, or 100). Uniformly sampled across trials. At 0% contrast (no visible stimulus), mice can still perform above chance using block probability prior.
intervals/trials	gabor_stimulus_side	Side on the screen where stimulus was presented: 'left' or 'right'. Even at 0% contrast (no visible stimulus), a side is assigned to the stimulus for the trial, enabling the mice to perform a correct response using prior knowledge of the block distribution.
intervals/trials	mouse_wheel_choice	Mouse's response: 'counter_clockwise' (counter_clockwise wheel turn moving stimulus leftward), 'clockwise' (clockwise wheel turn moving stimulus rightward), or 'none' (no response within 60s timeout). clockwise and counter_clockwise are wheel movements as seen from the perspective of the mouse
intervals/trials	is_mouse_rewarded	Whether the mouse received a water reward (True) or not (False). If False, an additional negative feedback consisting of an auditory white noise stimulus.
intervals/trials	reward_volume_uL	Volume of water reward in microliters (0 for incorrect/timeout trials).
intervals/trials	probability_left	Block prior probability for the visual stimulus to be presented on the left side of the screen. After an initial block of 90 unbiased trials (0.5), blocks alternate between a probability of 0.2 (right-biased) and 0.8 (left-biased). Block lengths: 20-100 trials from truncated geometric distribution (mean 51). Block changes are not cued.
intervals/trials	block_type	Block type based on stimulus probability bias: 'unbiased' (probability_left=0.5), 'left_block' (probability_left=0.8, stimulus 80% likely on the left side of the screen), or 'right_block' (probability_left=0.2, stimulus 80% likely on the right side of the screen).
processing/lick_times	(module)	Lick event timestamps detected from video-based tongue pose estimation. Licks are detected by tracking tongue landmark positions (Lightning Pose) and identifying frames with significant tongue movement. Licking indicates consummatory behavior following water reward delivery.

Location in NWB file	Name	Description (as written in the NWB file)
processing/lick_times	EventsLickTimes	Lick event timestamps detected from tongue pose estimation (Lightning Pose). Detection algorithm: frame-to-frame position changes in tongue landmarks (tongue_end_l_x, tongue_end_l_y, tongue_end_r_x, tongue_end_r_y) are computed, and frames where any coordinate changes by more than $\text{std}(\text{diff})/4$ are marked as lick events. If left and right camera data exist, the licks detected from both cameras are combined.
processing/motion_energy	(module)	Motion energy quantifies total movement within video regions of interest (ROIs). Values are computed as the sum of Euclidean pixel intensity differences between frames, normalized to [0, 1] range. Useful as a behavioral covariate for neural analysis.
processing/motion_energy	BodyCameraMotionEnergy	Motion energy calculated for a region of the body camera video. ROI dimensions: 173 pixels wide, 210 pixels tall, top-left corner at (115, 118). Calculation: For each frame, pixel intensity differences are computed between frame N and frame N+2 (default offset). The Euclidean norm (L2) of differences is summed across all ROI pixels, then min-max normalized to [0, 1] range. Higher values indicate more movement within the ROI. CAUTION: Video loading libraries may use different axis conventions. When loading with cv2 in Python, x and y axes are flipped. The region then becomes [118:328, 115:288].
processing/motion_energy	LeftCameraMotionEnergy	Motion energy calculated for a region of the left camera video. ROI dimensions: 121 pixels wide, 80 pixels tall, top-left corner at (380, 237). Calculation: For each frame, pixel intensity differences are computed between frame N and frame N+2 (default offset). The Euclidean norm (L2) of differences is summed across all ROI pixels, then min-max normalized to [0, 1] range. Higher values indicate more movement within the ROI. CAUTION: Video loading libraries may use different axis conventions. When loading with cv2 in Python, x and y axes are flipped. The region then becomes [237:317, 380:501].
processing/pose_estimation	(module)	Pose estimation from video using Lightning Pose.
processing/pose_estimation	BodyCamera	Estimated positions of body parts using Lightning Pose.
processing/pose_estimation	LeftCamera	Estimated positions of body parts using Lightning Pose.
processing/pose_estimation	Skeletons	
processing/pupil	(module)	Pupil diameter measurements from video-based eye tracking. Pupil diameter is estimated using pose estimation (DLC/Lightning Pose) to track keypoints around the pupil boundary. Diameter is computed as the median of multiple estimates (vertical, horizontal, and circular fits) for robustness. Pupil diameter correlates with arousal, attention, and cognitive state.
processing/pupil	LeftPupilDiameter	Raw pupil diameter estimated from pose tracking keypoints around the pupil boundary. Multiple diameter estimates are computed: vertical (top-bottom distance), horizontal (left-right distance), and circular fits from other keypoint pairs. The final value is the median of these estimates, providing robustness against individual tracking errors.
processing/pupil	LeftPupilDiameterSmoothed	Smoothed pupil diameter. This version has been temporally smoothed and interpolated over frames with missing or low-confidence keypoint detections. Useful for analyses where smooth trajectories are preferred over frame-by-frame accuracy.
processing/wheel	(module)	Rotary encoder wheel used for behavioral responses. The wheel (6.2 cm diameter) is positioned under the mouse's forepaws, and rotation is measured via a quadrature encoder (1024 ticks, X4 encoding = 4096 effective ticks per revolution). Mice turn the wheel to move visual stimuli and report perceptual decisions. Contains raw position, detected movements, and derived kinematics.
processing/wheel	WheelAccelerationSmoothed	Wheel angular acceleration derived from velocity (WheelVelocitySmoothed). Computed as the second derivative of the smoothed position signal.
processing/wheel	WheelPosition	Absolute unwrapped wheel angle recorded from a quadrature rotary encoder. The wheel (diameter 6.2 cm) is positioned under the mouse's forepaws and serves as the primary behavioral input device for reporting perceptual decisions. Sampling is event-driven: timestamps are recorded only when the wheel moves (i.e., when the encoder generates TTL edges), resulting in irregular inter-sample intervals. The encoder uses X4 decoding of two 90-degree phase-shifted channels, providing 4096 effective counts per revolution (angular resolution ~ 0.088 degrees or $2\pi/4096$ radians). Position is NOT periodic: values grow unboundedly as the wheel rotates, accumulating across multiple full revolutions (e.g., 3 full turns clockwise = -6π radians). The position is never wrapped back to $[0, 2\pi]$. Sign convention follows mathematical standard: counter-clockwise rotation (from the subject's perspective) is positive.
processing/wheel	WheelPositionSmoothed	Wheel position resampled to a uniform 1000 Hz grid and smoothed. The raw wheel position has irregular timestamps (event-driven from encoder edges). This series provides uniformly sampled position by: (1) linear interpolation to 1000 Hz, then (2) 8th order Butterworth lowpass filter (20 Hz corner, zero-phase) to remove high-frequency noise. This smoothed signal is used to derive velocity and acceleration via differentiation.
processing/wheel	WheelVelocitySmoothed	Wheel angular velocity derived from smoothed position (WheelPositionSmoothed). Computed as the first derivative of position after interpolation to 1000 Hz and lowpass filtering.
processing/wheel/WheelMovementIntervals	(table)	The onset and offset times of all detected movements. Movements are defined as a wheel movement of at least 0.012 rad over 200ms. For a rotary encoder of resolution 1024 in X4 encoding, this is equivalent to around 8 ticks. Movements below 50ms are discarded and two detected movements within 100ms of one another are considered as a single movement. For the onsets a lower threshold is used to find a more precise onset time. The wheel diameter is 6.2 cm and the number of ticks is 4096 per revolution.
processing/wheel/WheelMovementIntervals	start_time	Start time of epoch, in seconds
processing/wheel/WheelMovementIntervals	stop_time	Stop time of epoch, in seconds
processing/wheel/WheelMovementIntervals	peak_amplitude	The absolute maximum amplitude of each detected wheel movement, relative to onset position.

9. Exported files

File	Rows	Columns
tables/sessions.csv	459	session_id, session_uuid, asset_path, subject_id, lab, institution, sex, age_days, weight_kg, session_start_time, protocol, n_probes, n_trials, n_units, n_good_units, task_duration_s, regions
tables/units.csv	621,733	session_id, unit_id, unit_name, cluster_uuid, probe, region, region_name, ccf_x, ccf_y, ccf_z, distance_from_probe_tip_um, kilosort2_label, ibl_quality_score, good, peak_to_trough_ms, median_amplitude_uV, presence_ratio, isi_violations_ratio, spike_count, firing_rate_hz, task_rate_hz, prestim_rate_hz, stim_rate_hz, subject_id
tables/trials.csv	296,090	session_id, trial_id, start_time, stop_time, quiescence_period, gabor_stimulus_onset_time, auditory_cue_time, wheel_movement_onset_time, choice_registration_time, feedback_time, gabor_stimulus_offset_time, gabor_stimulus_contrast, gabor_stimulus_side, mouse_wheel_choice, is_mouse_rewarded, reward_volume_uL, probability_left, block_type, block_index
tables/region_summary.csv	540	region, region_name, n_units, n_sessions, n_mice, firing_rate_mean_hz, firing_rate_sd_hz, firing_rate_median_hz, prestim_rate_mean_hz, prestim_rate_median_hz, stim_rate_mean_hz, peak_to_trough_ms_median
tables/dataset_metadata.json, nwb_file_metadata.json, nwb_descriptions.csv		the published metadata quoted in sections 1 and 8
trial_unit_counts.parquet	trials × units per session	session_id, trial_id, unit_id, count_prestim, count_stim, count_remove, count_feedback